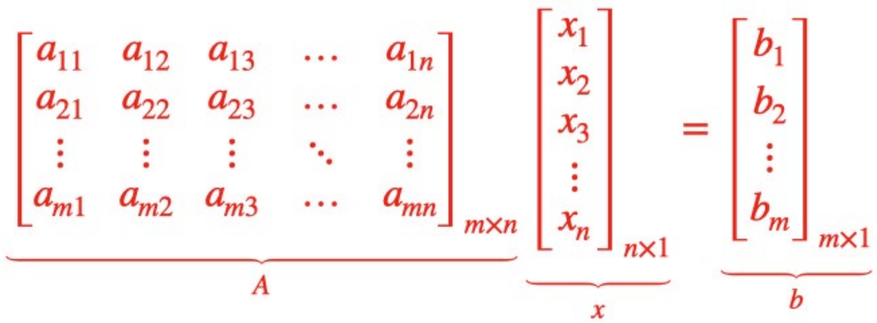


CX1104: **Linear Algebra for Computing**

Chap. No : **7.1.1**

Lecture : **Least Squares**

Topic : **Introduction**

Concept : **Consistency in a System of Equations**

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su**, **Vishal Choudhari**

Rev: 30th June 2020 Last updated: 10 Dec 2025

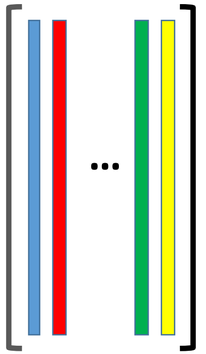
Consistency in a System of Equations

Consider solving the system of equations: $Ax = b$

Note:

- Matrix $A \in R^{M \times N}$, where
  - M denotes no. of rows/equations
  - N denotes no. of columns/unknowns
- $x \in R^N$
- $b \in R^M$
- The above system of equations can either be
  1. consistent (or)
  2. inconsistent.

Based on M & N, there exist three cases:



$M \gg N$

More equations, less unknowns.

Hence, **over-determined**! **Typically** this will result in inconsistent system of equations



$M \approx N$



$M \ll N$

Less equations, more unknowns.

Hence, **under-determined**! **Typically,** this will result in infinite solutions

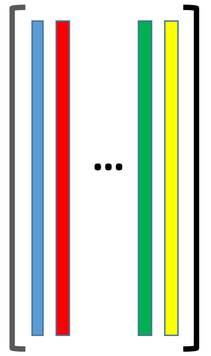
Consistency in a System of Equations

$$Ax = b$$

Consistent (solutions exist)

- b is in column space of A, i.e,
  b is formed by linear combinations of A's columns.
- Rank (A) = Rank ($A|b$), i.e,
  rank of A is same as that of the augmented matrix.

Inconsistent (solutions don't exist)



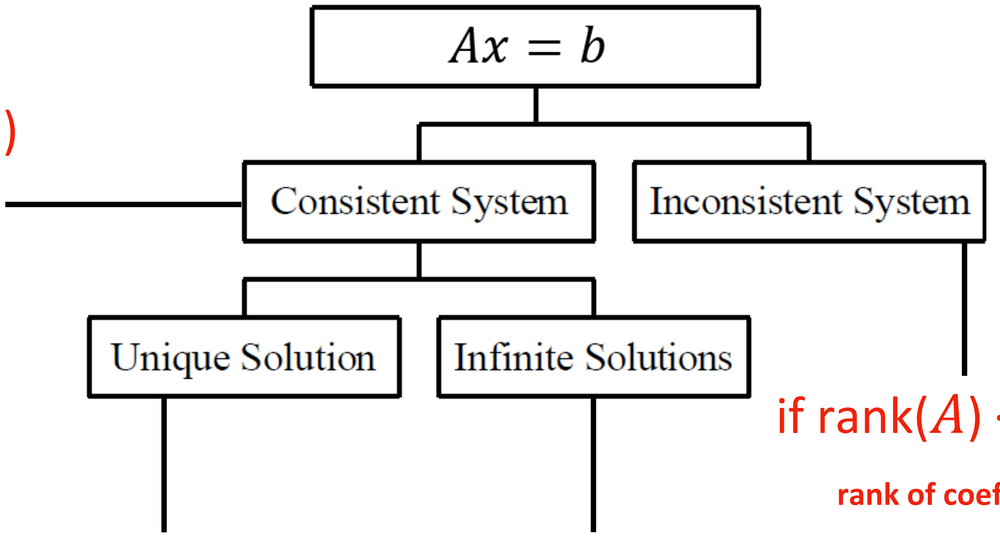
$M \gg N$

- b is NOT in column space of A, i.e,
  b is NOT formed by linear combinations of A's columns.
- **Typically** occurs when $M \gg N$ (**over-determined**), i.e,
  there exist more equations than unknowns.
- The rows of A are dependent but,
  their corresponding b values are not consistent.
- Rank (A) < Rank ($A|b$), i.e,
  rank of A is less than that of the augmented matrix.

Consistency in a System of Equations

A system of equations can be consistent or inconsistent. What does that mean?

A system of equations $Ax = b$ is consistent if there is a solution, and it is inconsistent if there is no solution. However, consistent system of equations does not mean a unique solution, that is, a consistent system of equation may have a unique solution or infinite solutions.



if rank(A) = rank($A|b$)

rank of coefficient matrix = rank of augmented matrix

if rank(A) < rank($A|b$)

rank of coefficient matrix < rank of augmented matrix

if rank(A) = N

rank of coefficient matrix = rank of augmented matrix = no. of unknowns

if rank(A) < N

rank of coefficient matrix = rank of augmented matrix < no. of unknowns

NOTE: Rank (A) is the maximum number of independent rows or columns of A.

You can find number of independent row or columns by:
1. row reduction process
2. rank(A) in MATLAB

Ref: https://en.wikipedia.org/wiki/Augmented_matrix
Ref: https://www.mathsisfun.com/algebra/matrix-rank.html

Note: rank (A) > rank ($A|b$) is never possible. Why?

Examples

$\text{rank}(A) = \text{rank}(A|b) = N$

Consistent and Unique Solution

a) The system of equations

$$\begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

```
>> A_b = [ 2 4 6; 1 3 4]

A_b =

     2     4     6
     1     3     4

>> rank(A_b)

ans =

     2
```

is a consistent system of equations as it has a unique solution, that is,

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Inconsistent and No solutions Exist

c) The system of equations

$$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

```
>> A = [2 4; 1 2]

A =

     2     4
     1     2

>> rank(A)

ans =

     1
```

```
>> A_b = [ 2 4 6; 1 2 4]

A_b =

     2     4     6
     1     2     4

>> rank(A_b)

ans =

     2
```

$\text{rank}(A) < \text{rank}(A|b)$

Consistent and Having Infinite Solutions

b) The system of equations

$$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

```
>> A_b = [ 2 4 6; 1 2 3]

A_b =

     2     4     6
     1     2     3

>> rank(A_b)

ans =

     1
```

is also a consistent system of equations but it has infinite solutions as given as follows.

Expanding the above set of equations,

$$2x + 4y = 6$$
$$x + 2y = 3$$

you can see that they are the same equation. Hence any combination of (x, y) that satisfies

$$2x + 4y = 6$$

is a solution. For example $(x, y) = (1,1)$ is a solution and other solutions include $(x, y) = (0.5,1.25)$, $(x, y) = (0,\ 1.5)$ and so on.

$\text{rank}(A) = \text{rank}(A|b) < N$

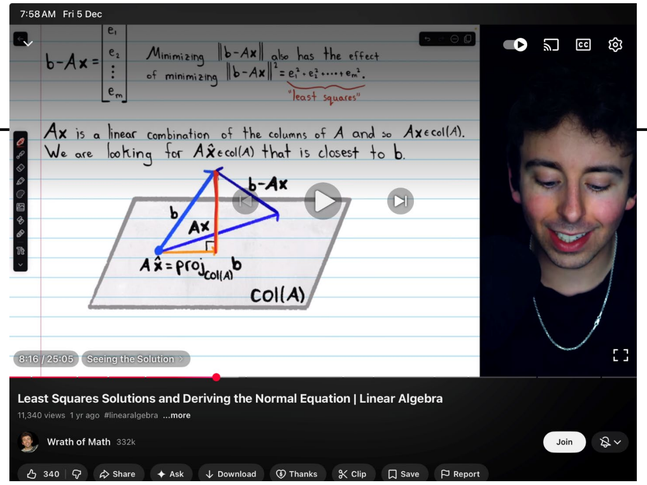
Ref: Introduction to Matrix Algebra, Autar Kaw (pg 68)

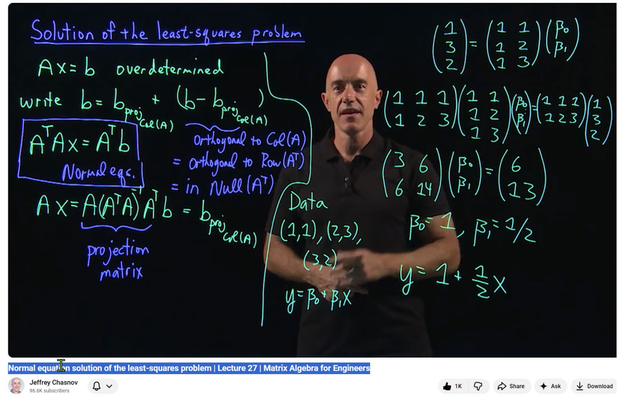
References for Least Squares

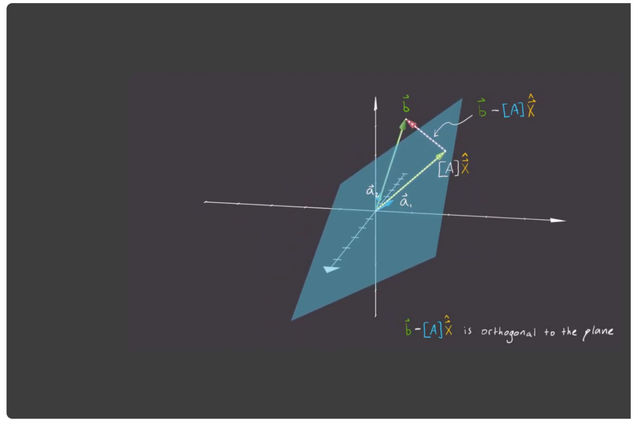
1) YouTube, Wrath of Math, “Least Squares Solutions and Deriving the Normal Equation”
https://www.youtube.com/watch?v=9o5a9oBhVBk

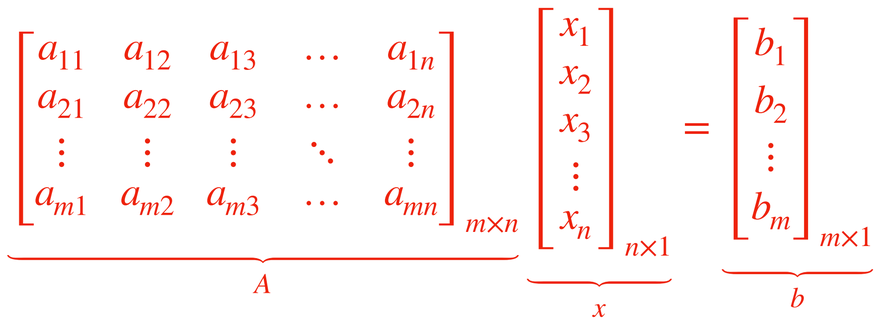
2) YouTube, Chasnov, “**Normal equation solution of the least-squares problem | Lecture 27 | Matrix Algebra for Engineers**“ https://www.youtube.com/watch?v=WABC6wmuLOk

3) YouTube, Virtually Passed, “**Least Squares Formula PROOF**“,
https://www.youtube.com/watch?v=wJ35udCsHVc









CX1104: **Linear Algebra for Computing**

Chap. No : **7.1.2**

Lecture : **Least Squares**

Topic : **Introduction**

Concept : **The Least Squares Problem**

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su**, **Vishal Choudhari**

Rev: 2nd July 2020

Consistency in a System of Equations

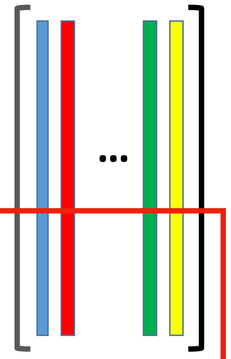
$$Ax = b$$

Consistent
(solutions exist)

- b is in column space of A, i.e,
  b is formed by linear combinations of A's columns.
- Rank (A) = Rank ($A \mid b$), i.e,
  rank of A is same as that of the augmented matrix.

Consider

Inconsistent
(solutions don't exist)



$M \gg N$

- b is NOT in column space of A, i.e,
  b is NOT formed by linear combinations of A's columns.
- Occurs when $M \gg N$ (**over-determined**), i.e,
  there exist more equations than unknowns.
- The rows of A are dependent but,
  their corresponding b values are not consistent.
- Rank (A) < Rank ($A \mid b$), i.e,
  rank of A is less than that of the augmented matrix.

Least Squares Solution for Inconsistent Equations

Consider solving the system of equations: $Ax = b$

Note:

- Matrix $A \in R^{M \times N}$, where
  - M denotes no. of rows/equations
  - N denotes no. of columns/unknowns
- $x \in R^N$
- $b \in R^M$
- When $M \gg N$,
  - the system is over-determined
  - the equations may be inconsistent
  - there may be no solution

Best we can do?

Find x such that Ax is as close to b as possible!

If A is $m \times n$ and $\mathbf{b}$ is in $\mathbb{R}^m$, a **least-squares solution** of $A\mathbf{x} = \mathbf{b}$ is an $\hat{\mathbf{x}}$ in $\mathbb{R}^n$ such that

$$\|\mathbf{b} - A\hat{\mathbf{x}}\| \leq \|\mathbf{b} - A\mathbf{x}\|$$

for all $\mathbf{x}$ in $\mathbb{R}^n$.

Think of $A\mathbf{x}$ as an *approximation* to $\mathbf{b}$. The smaller the distance between $\mathbf{b}$ and $A\mathbf{x}$, given by $\|\mathbf{b} - A\mathbf{x}\|$, the better the approximation. The **general least-squares problem** is to find an $\mathbf{x}$ that makes $\|\mathbf{b} - A\mathbf{x}\|$ as small as possible. The adjective "least-squares" arises from the fact that $\|\mathbf{b} - A\mathbf{x}\|$ is the square root of a sum of squares.

Lay, Linear Algebra and its Applications (4th Edition)

360 CHAPTER 6 Orthogonality and Least Squares

Definitions

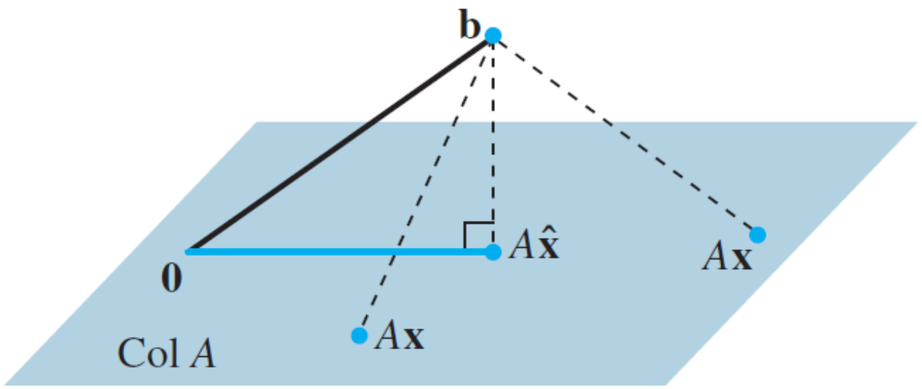


FIGURE 1 The vector **b** is closer to $A\hat{\mathbf{x}}$ than to $A\mathbf{x}$ for other **x**.

The most important aspect of the least-squares problem is that no matter what **x** we select, the vector $A\mathbf{x}$ will necessarily be in the column space, Col A. So we seek an **x** that makes $A\mathbf{x}$ the closest point in Col A to **b**. See Fig. 1. (Of course, if **b** happens to be in Col A, then **b** *is* $A\mathbf{x}$ for some **x**, and such an **x** is a "least-squares solution.")

If a linear system is consistent, then its exact solutions are the same as its least squares solutions, in which case the least squares error is zero.

NOTE:

When the linear system $Ax = b$ is inconsistent, b does not lie in the column space of A.

To explain the terminology in this problem, suppose that the column form of $\mathbf{b} - A\mathbf{x}$ is

$$\mathbf{b} - A\mathbf{x} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_m \end{bmatrix}$$

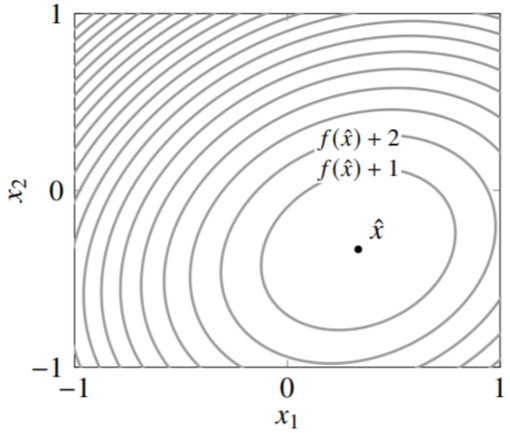
The term "least squares solution" results from the fact that minimizing $\|\mathbf{b} - A\mathbf{x}\|$ also has the effect of minimizing $\|\mathbf{b} - A\mathbf{x}\|^2 = e_1^2 + e_2^2 + \cdots + e_m^2$.

Lay, Linear Algebra and its Applications (4th Edition)

Example

Example

$$A = \begin{bmatrix} 2 & 0 \\ -1 & 1 \\ 0 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$



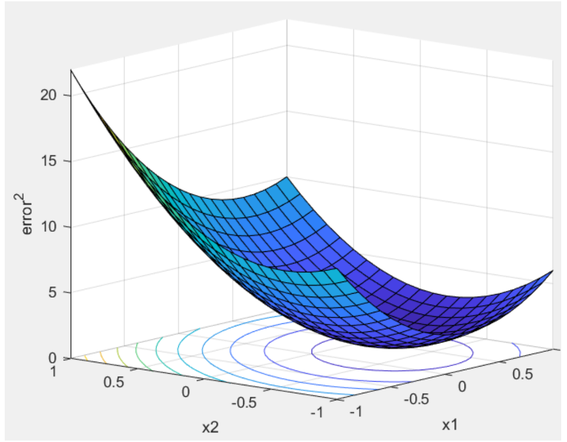
- the least squares solution $\hat{x}$ minimizes

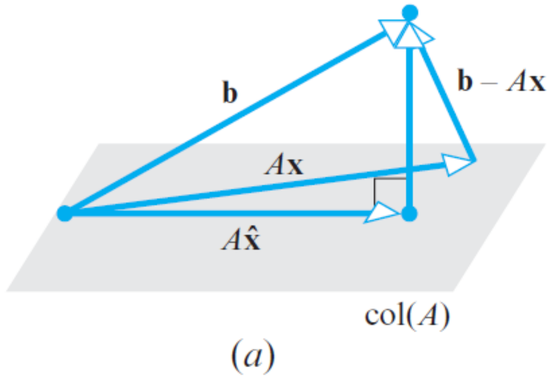
$$f(x) = \|Ax - b\|^2 = (2x_1 - 1)^2 + (-x_1 + x_2)^2 + (2x_2 + 1)^2$$

- to find $\hat{x}$, set derivatives with respect to x_1 and x_2 equal to zero:

$$10x_1 - 2x_2 - 4 = 0, \qquad -2x_1 + 10x_2 + 4 = 0$$

solution is $(\hat{x}_1, \hat{x}_2) = (1/3, -1/3)$





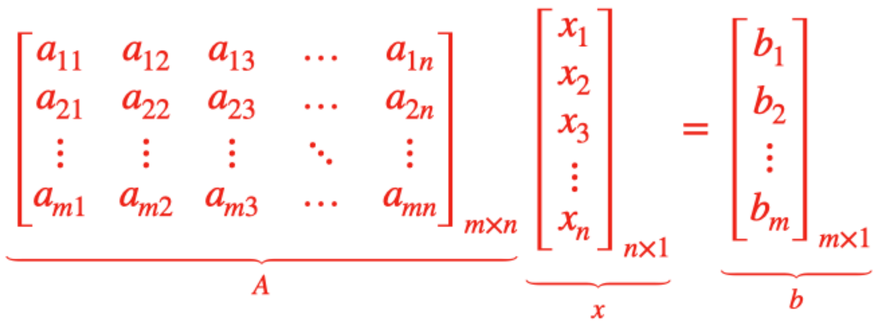

```
opStr =

    'x1=0.30,  x2=-0.30,  err^2=0.680 '
```

```
%ch6_4_Ex1.m
%Chng Eng Siong, plotting the error wrt x
close all; clear all;
A = [2 0; -1 1; 0 2];
b = [1 0 -1]';
[x1,x2] = meshgrid(-1:0.1:1, -1:0.1:1);
[m,n] = size(x1);
z = zeros(m,n);
for i=1:m
    for j=1:n
        z(i,j) = norm(b - (x1(i,j)*A(:,1)+x2(i,j)*A(:,2))).^2;
    end
end
surfc(x1,x2,z)
xlabel('x1'); ylabel('x2'); zlabel('error^2');

% Lets print the min value and the x vector
minIdx = find(z == min(z(:)));
x1(minIdx), x2(minIdx), z(minIdx)
opStr = sprintf('x1=%0.2f,  x2=%0.2f,  err^2=%.3f ',x1(minIdx),x2(minIdx),z(minIdx))
```

Boyd – Lecture 6
Ref: https://www.youtube.com/watch?v=ZKhZclqfR5E&t=1850s



CX1104: **Linear Algebra for Computing**

Chap. No : **7.1.3**

Lecture : **Least Squares**

Topic : **Solving the Least Squares Problem**

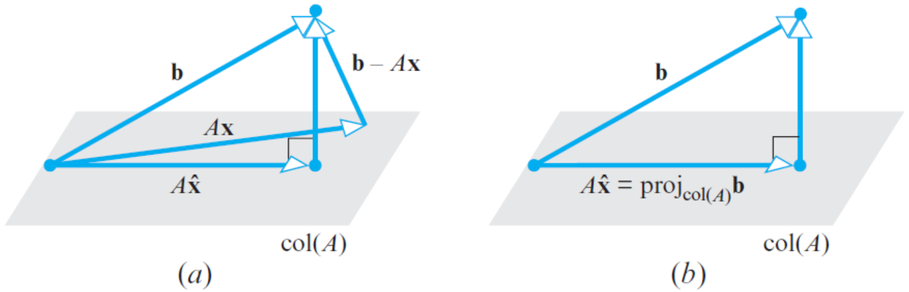
Concept : **Best Approx. Theorem and Normal Equation**

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su**, **Vishal Choudhari**

Rev: 2nd July 2020

Best Approximation Theorem



THEOREM 6.4.1 Best Approximation Theorem

If W is a finite-dimensional subspace of an inner product space V, and if $\mathbf{b}$ is a vector in V, then $\text{proj}_W \mathbf{b}$ *is the* ***best approximation*** *to* $\mathbf{b}$ *from W in the sense that*

$$\|\mathbf{b} - \text{proj}_W \mathbf{b}\| < \|\mathbf{b} - \mathbf{w}\|$$

for every vector $\mathbf{w}$ *in W that is different from* $\text{proj}_W \mathbf{b}$.

Anton:

Proof For every vector $\mathbf{w}$ in W, we can write

$$\mathbf{b} - \mathbf{w} = (\mathbf{b} - \text{proj}_W \mathbf{b}) + (\text{proj}_W \mathbf{b} - \mathbf{w})$$

But $\text{proj}_W \mathbf{b} - \mathbf{w}$, being a difference of vectors in W, is itself in W; and since $\mathbf{b} - \text{proj}_W \mathbf{b}$ is orthogonal to W, the two terms on the right side of (1) are orthogonal. Thus, it follows from the Theorem of Pythagoras (Theorem 6.2.3) that

$$\|\mathbf{b} - \mathbf{w}\|^2 = \|\mathbf{b} - \text{proj}_W \mathbf{b}\|^2 + \|\text{proj}_W \mathbf{b} - \mathbf{w}\|^2$$

If $\mathbf{w} \neq \text{proj}_W \mathbf{b}$, it follows that the second term in this sum is positive, and hence that

$$\|\mathbf{b} - \text{proj}_W \mathbf{b}\|^2 < \|\mathbf{b} - \mathbf{w}\|^2$$

Since norms are nonnegative, it follows (from a property of inequalities) that

$$\|\mathbf{b} - \text{proj}_W \mathbf{b}\| < \|\mathbf{b} - \mathbf{w}\| \quad \blacktriangleleft$$

The Normal Equation

$$Ax = b$$

Multiplying both sides by A^T

$$A^TAx = A^Tb$$

Normal Equation!

The set of least-squares solutions of $A\mathbf{x} = \mathbf{b}$ coincides with the nonempty set of solutions of the normal equations $A^TA\mathbf{x} = A^T\mathbf{b}$.

Proved in the next slide!

Why called "Normal"?
Ref: https://mathworld.wolfram.com/NormalEquation.html

Solution of the General Least-Squares Problem

Given A and $\mathbf{b}$ as above, apply the Best Approximation Theorem in Section 6.3 to the subspace Col A. Let

$$\hat{\mathbf{b}} = \text{proj}_{\text{Col } A}\,\mathbf{b}$$

Because $\hat{\mathbf{b}}$ is in the column space of A, the equation $A\mathbf{x} = \hat{\mathbf{b}}$ *is* consistent, and there is an $\hat{\mathbf{x}}$ in $\mathbb{R}^n$ such that

$$A\hat{\mathbf{x}} = \hat{\mathbf{b}} \tag{1}$$

Since $\hat{\mathbf{b}}$ is the closest point in Col A to $\mathbf{b}$, a vector $\hat{\mathbf{x}}$ is a least-squares solution of $A\mathbf{x} = \mathbf{b}$ if and only if $\hat{\mathbf{x}}$ satisfies (1). Such an $\hat{\mathbf{x}}$ in $\mathbb{R}^n$ is a list of weights that will build $\hat{\mathbf{b}}$ out of the columns of A. See Fig. 2. [There are many solutions of (1) if the equation has free variables.]

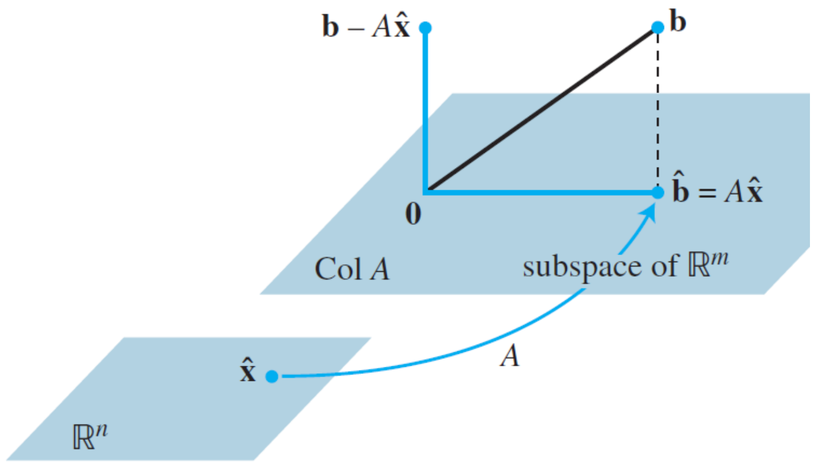


FIGURE 2 The least-squares solution $\hat{\mathbf{x}}$ is in $\mathbb{R}^n$.

Lay, Linear Algebra and its Applications (4th Edition)

The Normal Equation Proof

Suppose $\hat{\mathbf{x}}$ satisfies $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$. By the Orthogonal Decomposition Theorem in Section 6.3, the projection $\hat{\mathbf{b}}$ has the property that $\mathbf{b} - \hat{\mathbf{b}}$ is orthogonal to Col A, so $\mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to each column of A. If $\mathbf{a}_j$ is any column of A, then $\mathbf{a}_j \cdot (\mathbf{b} - A\hat{\mathbf{x}}) = 0$, and $\mathbf{a}_j^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0$. Since each $\mathbf{a}_j^T$ is a row of A^T,

$$A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0} \tag{2}$$

(This equation also follows from Theorem 3 in Section 6.1.) Thus

$$A^T\mathbf{b} - A^TA\hat{\mathbf{x}} = \mathbf{0}$$
$$A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$$

These calculations show that each least-squares solution of $A\mathbf{x} = \mathbf{b}$ satisfies the equation

$$A^TA\mathbf{x} = A^T\mathbf{b} \tag{3}$$

The matrix equation (3) represents a system of equations called the **normal equations** for $A\mathbf{x} = \mathbf{b}$. A solution of (3) is often denoted by $\hat{\mathbf{x}}$.

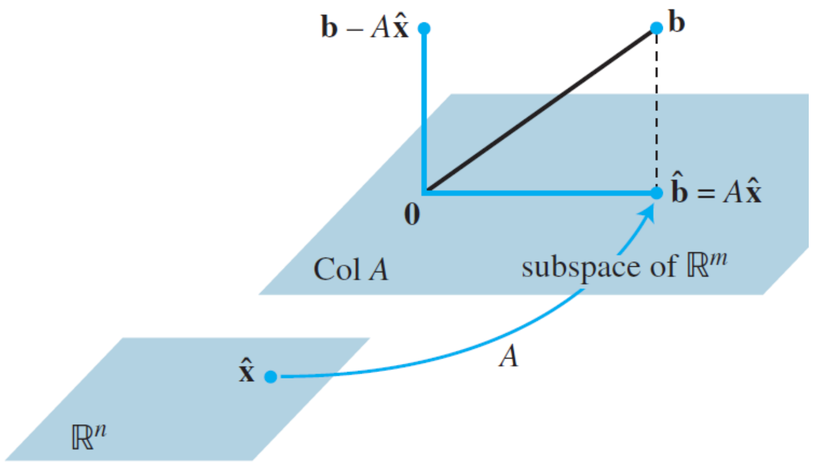


FIGURE 2 The least-squares solution $\hat{\mathbf{x}}$ is in $\mathbb{R}^n$.

THEOREM 14

Let A be an $m \times n$ matrix. The following statements are logically equivalent:

a. The equation $A\mathbf{x} = \mathbf{b}$ has a unique least-squares solution for each $\mathbf{b}$ in $\mathbb{R}^m$.
b. The columns of A are linearly indpendent.
c. The matrix A^TA is invertible.

When these statements are true, the least-squares solution $\hat{\mathbf{x}}$ is given by

$$\hat{\mathbf{x}} = (A^TA)^{-1}A^T\mathbf{b} \tag{4}$$

Examples

EXAMPLE 1 Find a least-squares solution of the inconsistent system $A\mathbf{x} = \mathbf{b}$ for

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$$

SOLUTION To use normal equations (3), compute:

$$A^TA = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix}$$

$$A^T\mathbf{b} = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

Then the equation $A^TA\mathbf{x} = A^T\mathbf{b}$ becomes

$$\begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

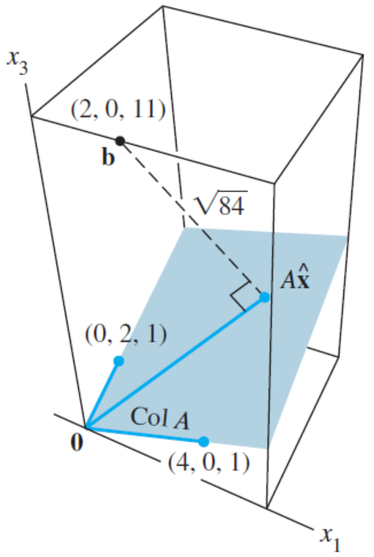


FIGURE 3

Row operations can be used to solve this system, but since A^TA is invertible and 2×2, it is probably faster to compute

$$(A^TA)^{-1} = \frac{1}{84}\begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix}$$

and then to solve $A^TA\mathbf{x} = A^T\mathbf{b}$ as

$$\begin{aligned} \hat{\mathbf{x}} &= (A^TA)^{-1}A^T\mathbf{b} \\ &= \frac{1}{84}\begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix}\begin{bmatrix} 19 \\ 11 \end{bmatrix} = \frac{1}{84}\begin{bmatrix} 84 \\ 168 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \end{aligned}$$

In many calculations, A^TA is invertible, but this is not always the case. The next

Examples

EXAMPLE 3 Given A and $\mathbf{b}$ as in Example 1, determine the least-squares error in the least-squares solution of $A\mathbf{x} = \mathbf{b}$.

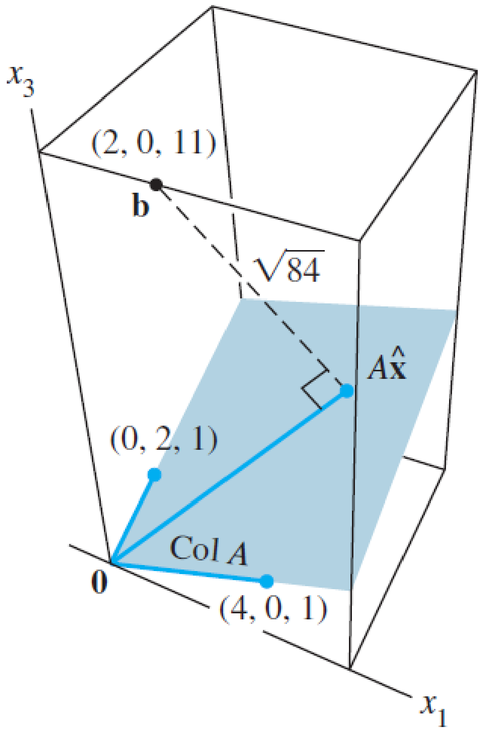


FIGURE 3

SOLUTION From Example 1,

$$\mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} \quad \text{and} \quad A\hat{\mathbf{x}} = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 3 \end{bmatrix}$$

Hence

$$\mathbf{b} - A\hat{\mathbf{x}} = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} - \begin{bmatrix} 4 \\ 4 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \\ 8 \end{bmatrix}$$

and

$$\|\mathbf{b} - A\hat{\mathbf{x}}\| = \sqrt{(-2)^2 + (-4)^2 + 8^2} = \sqrt{84}$$

The least-squares error is $\sqrt{84}$. For any $\mathbf{x}$ in $\mathbb{R}^2$, the distance between $\mathbf{b}$ and the vector $A\mathbf{x}$ is at least $\sqrt{84}$. See Fig. 3. Note that the least-squares solution $\hat{\mathbf{x}}$ itself does not appear in the figure. ■

Examples

EXAMPLE 2 Find a least-squares solution of $A\mathbf{x} = \mathbf{b}$ for

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -3 \\ -1 \\ 0 \\ 2 \\ 5 \\ 1 \end{bmatrix}$$

Note the linear dependency in the rows and columns of A:

- Column 1 = Column 2 + Column 3 + Column 4
- Rows 1 & 2 are same, but their corresponding b values are different (inconsistent)
- Rows 3 & 4 are same, but their corresponding b values are different (inconsistent)
- Rows 5 & 6 are same, but their corresponding b values are different (inconsistent)

SOLUTION Compute

$$A^TA = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 2 & 2 & 2 \\ 2 & 2 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 2 & 0 & 0 & 2 \end{bmatrix}$$

A^TA

Note that A^TA is always a square matrix.

$$A^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -3 \\ -1 \\ 0 \\ 2 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \\ 2 \\ 6 \end{bmatrix}$$

Lay, Linear Algebra and its Applications (4th Edition)

6.5 Least-Squares Problems 361

The augmented matrix for $A^TA\mathbf{x} = A^T\mathbf{b}$ is

Reduced to

$$\underbrace{\begin{bmatrix} 6 & 2 & 2 & 2 \\ 2 & 2 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 2 & 0 & 0 & 2 \end{bmatrix}}_{A^TA} \underbrace{\begin{bmatrix} 4 \\ -4 \\ 2 \\ 6 \end{bmatrix}}_{A^Tb} \sim \begin{bmatrix} 1 & 0 & 0 & 1 & 3 \\ 0 & 1 & 0 & -1 & -5 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The general solution is $x_1 = 3 - x_4$, $x_2 = -5 + x_4$, $x_3 = -2 + x_4$, and x_4 is free. So the general least-squares solution of $A\mathbf{x} = \mathbf{b}$ has the form

$$\hat{\mathbf{x}} = \begin{bmatrix} 3 \\ -5 \\ -2 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

■

Note: Here, there are infinitely many solutions with the same least square error.
Note: Here, A^TA is not invertible (its determinant is 0).

A^TA may not be invertible if:

- some columns are linearly dependent (i.e. we have redundant features) (**as in this example**)
  - solution: remove the linear dependency
- too many features ($m < n$)
  - solution: delete some features, there are too many features for the amount of data we have

Ref: http://mlwiki.org/index.php/Normal_Equation

Ref: Andrew Ng discussing this phenomenon-
https://www.coursera.org/lecture/machine-learning/normal-equation-noninvertibility-zSiE6

Examples

```
% pg 362 Lay's book, Example 2 - Least Squares, when A'A is singular
close all; clear all;
A = [1 1 0 0; 1 1 0 0; 1 0 1 0; 1 0 1 0; 1 0 0 1; 1 0 0 1];
b = [-3 -1 0 2 5 1]';

AtA = A'*A
rank_Ata = rank(AtA)  % A'*A is singular, we check its rank

% Ax = b;
x1 = pinv(A)*b
x2 = inv(A'*A)*A'*b   % This is what we think we should do
% compare inv(A'*A) bs pinv(A'*A)
disp("using normal inverser (A'*A):");
inv(A'*A)
disp("using pinverser (A'*A):");
pinv(A'*A)

x3 = pinv(A'*A)*A'*b  % This is what Andy Ng suggest to d
```

```
AtA =

     6     2     2     2
     2     2     0     0
     2     0     2     0
     2     0     0     2
```

```
rank_Ata =

     3
```

```
x1 =

    0.5000
   -2.5000
    0.5000
    2.5000
```

```
x2 =

    1
   -3
    0
    2
```

```
Warning: Matrix is close to singular or badly scaled.
> In Lay_example2_pg362 (line 8)
```

$A^T A$

```
ans =

   1.0e+15 *

    1.5012   -1.5012   -1.5012   -1.5012
   -1.5012    1.5012    1.5012    1.5012
   -1.5012    1.5012    1.5012    1.5012
   -1.5012    1.5012    1.5012    1.5012
```

$A^T A$ is non-invertible. Hence MATLAB computes its inverse as a very large value $\Longrightarrow \infty$

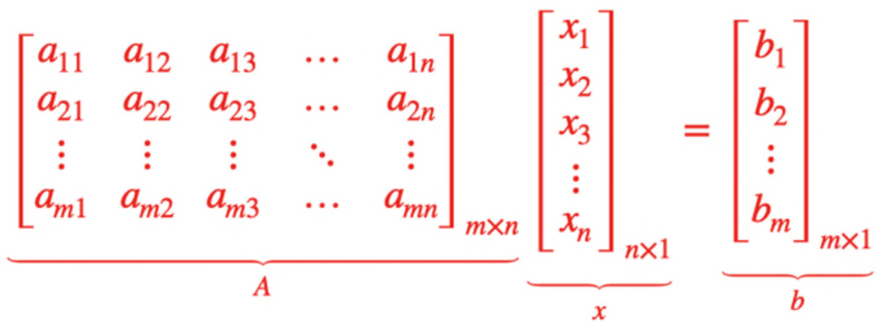
```
ans =

    0.0938    0.0312    0.0313    0.0313
    0.0313    0.3437   -0.1562   -0.1563
    0.0312   -0.1562    0.3438   -0.1562
    0.0313   -0.1562   -0.1563    0.3438
```

```
x3 =

    0.5000
   -2.5000
    0.5000
    2.5000
```

NOTE: Pseudo-inverse (pinv) will be introduced later.



CX1104: **Linear Algebra for Computing**

Chap. No : **7.1.3A**

Lecture : **Least Squares**

Topic : **Least Squares**

Concept : **Supplementary –** Proving Solutions of Normal Equation are Least Square Solutions

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su**, **Vishal Choudhari**

Revised: 18 Dec 2024

Rev: 2nd July 2020

Proving Solutions of Normal Equation are Least Square Solutions

The set of least-squares solutions of $A\mathbf{x} = \mathbf{b}$ coincides with the nonempty set of solutions of the normal equations $A^TA\mathbf{x} = A^T\mathbf{b}$.

Proofs by:

1. Gilbert Strang, MIT (2 Proofs)
2. Jeffrey Chasnov, HKUST
3. Alex Townsend, MIT
4. Alexey Grigorev
5. Quora

Gilbert Strang (Proof 1)

In Strang's Lecture 15 video, Strang: 18.06

Projection onto Subspace

https://youtu.be/Y_Ac6KiQ1t0?t=951

1. 3-D, projecting a point b onto a plane span by A's columns (2-D) (18:50)
   - Vector b is not in the plane (span by A's columns). (Fig a)
   - What is that plane? (19:40)
     - The plane is span by columns of A: a_1 and a_2 and hence, form the basis for the subspace.
     - The columns of A are independent, and need NOT be perpendicular.
   - (22:20) $e = b - p$ (is orthogonal to the plane of A) (crucial fact)

2. What is small p (the projected vector of b on subspace spanned by A's columns)?
   - 23:45: p is some linear combination of columns of A. (Fig b)
   - $p = A\hat{x}$, we are looking for $\hat{x}$ such that error vector e is orthogonal to plane.

3. Now e is orthogonal to the column space of A. (26:00)
   - We have 2 equations to solve for 2 unknowns (fig c)
   - Reducing it to $A^T(b - A\hat{x}) = 0$ (fig d) -> The normal equation

Fig (a)

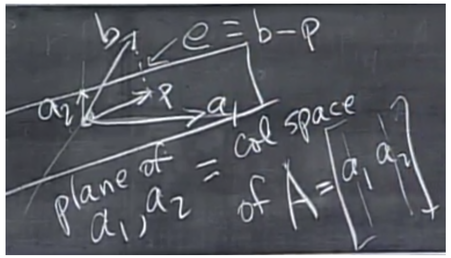


Fig (b)

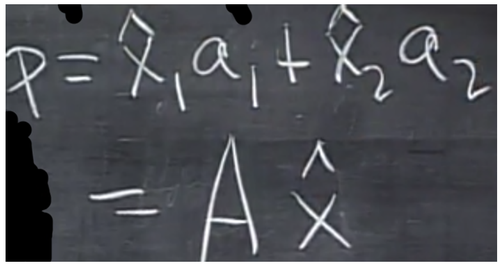


Fig (c)

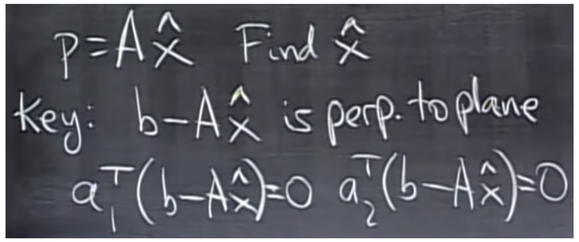
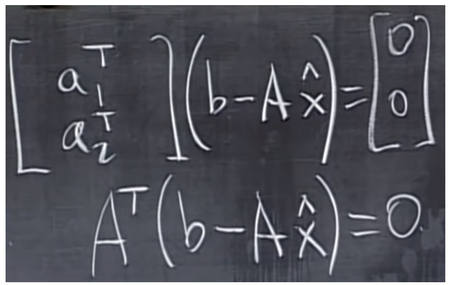
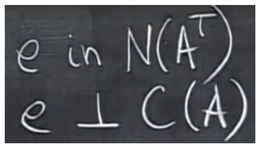


Fig (d)





Gilbert Strang (Proof 2 - Using Dot Product)

Projections onto subspaces

Projections

If we have a vector $\mathbf{b}$ and a line determined by a vector $\mathbf{a}$, how do we find t
point on the line that is closest to $\mathbf{b}$?

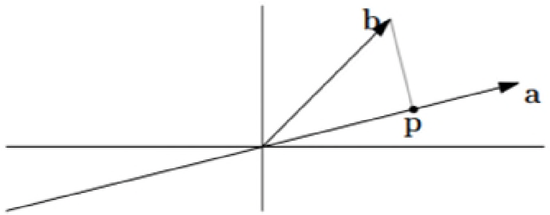


Figure 1: The point closest to $\mathbf{b}$ on the line determined by $\mathbf{a}$.

We can see from Figure 1 that this closest point $\mathbf{p}$ is at the intersecti
formed by a line through $\mathbf{b}$ that is orthogonal to $\mathbf{a}$. If we think of $\mathbf{p}$ as
approximation of $\mathbf{b}$, then the length of $\mathbf{e} = \mathbf{b} - \mathbf{p}$ is the error in that appro
mation.

We could try to find $\mathbf{p}$ using trigonometry or calculus, but it's easier to u
linear algebra. Since $\mathbf{p}$ lies on the line through $\mathbf{a}$, we know $\mathbf{p} = x\mathbf{a}$ for so
number x. We also know that $\mathbf{a}$ is perpendicular to $\mathbf{e} = \mathbf{b} - x\mathbf{a}$:

$$\begin{aligned} \mathbf{a}^T(\mathbf{b} - x\mathbf{a}) &= 0 \\ x\mathbf{a}^T\mathbf{a} &= \mathbf{a}^T\mathbf{b} \\ x &= \frac{\mathbf{a}^T\mathbf{b}}{\mathbf{a}^T\mathbf{a}}, \end{aligned}$$

and $\mathbf{p} = \mathbf{a}x = \mathbf{a}\frac{\mathbf{a}^T\mathbf{b}}{\mathbf{a}^T\mathbf{a}}$. Doubling $\mathbf{b}$ doubles $\mathbf{p}$. Doubling $\mathbf{a}$ does not affect $\mathbf{p}$.

Ref: Lecture Writeup
https://ocw.mit.edu/courses/mathematics/18-06sc-linear-algebra-fall-2011/least-squares-determinants-and-eigenvalues/projections-onto-subspaces/MIT18_06SCF11_Ses2.2sum.pdf

Why project?

As we know, the equation $A\mathbf{x} = \mathbf{b}$ may have no solution. The vector $A\mathbf{x}$ is always in the column space of A, and $\mathbf{b}$ is unlikely to be in the column space. So, we project $\mathbf{b}$ onto a vector $\mathbf{p}$ in the column space of A and solve $A\hat{\mathbf{x}} = \mathbf{p}$.

Projection in higher dimensions

In $\mathbb{R}^3$, how do we project a vector $\mathbf{b}$ onto the closest point $\mathbf{p}$ in a plane?

If $\mathbf{a}_1$ and $\mathbf{a}_2$ form a basis for the plane, then that plane is the column space of the matrix $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 \end{bmatrix}$.

We know that $\mathbf{p} = \hat{x}_1\mathbf{a}_1 + \hat{x}_2\mathbf{a}_2 = A\hat{\mathbf{x}}$. We want to find $\hat{\mathbf{x}}$. There are many ways to show that $\mathbf{e} = \mathbf{b} - \mathbf{p} = \mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to the plane we're projecting onto, after which we can use the fact that $\mathbf{e}$ is perpendicular to $\mathbf{a}_1$ and $\mathbf{a}_2$:

$$\mathbf{a}_1^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0 \quad \text{and} \quad \mathbf{a}_2^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0.$$

In matrix form, $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$. When we were projecting onto a line, A only had one column and so this equation looked like: $a^T(\mathbf{b} - x\mathbf{a}) = \mathbf{0}$.

Note that $\mathbf{e} = \mathbf{b} - A\hat{\mathbf{x}}$ is in the nullspace of A^T and so is in the left nullspace of A. We know that everything in the left nullspace of A is perpendicular to the column space of A, so this is another confirmation that our calculations are correct.

We can rewrite the equation $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$ as:

$$A^TA\hat{\mathbf{x}} = A^T\mathbf{b}.$$

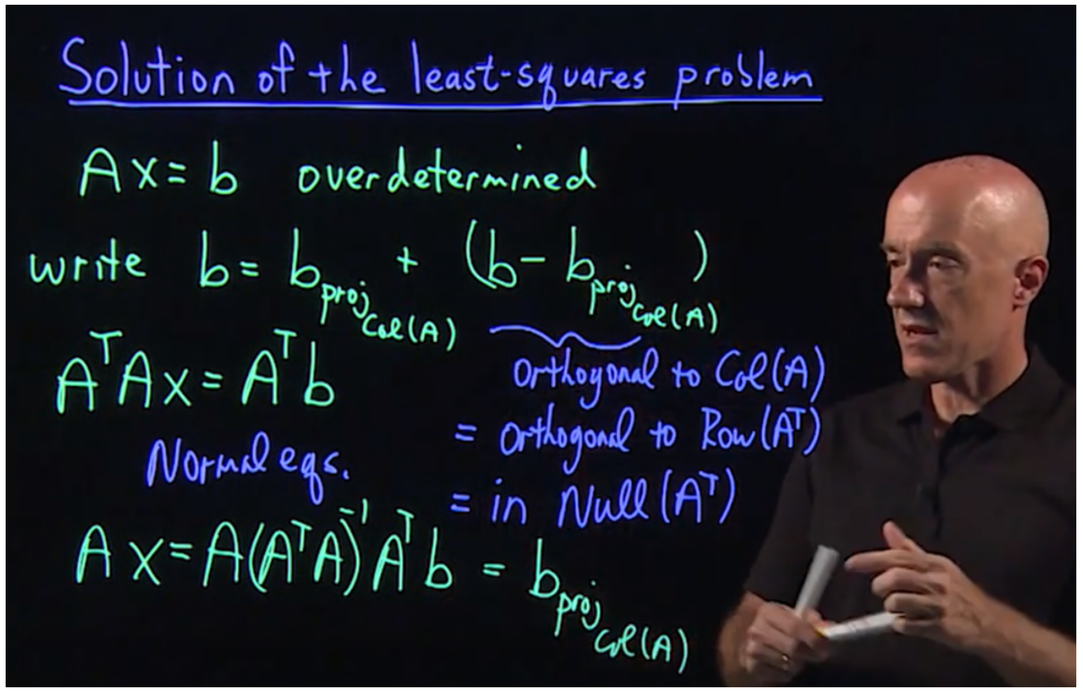
When projecting onto a line, A^TA was just a number; now it is a square matrix. So instead of dividing by $\mathbf{a}^T\mathbf{a}$ we now have to multiply by $(A^TA)^{-1}$

In n dimensions,

$$\begin{aligned} \hat{\mathbf{x}} &= (A^TA)^{-1}A^T\mathbf{b} \\ \mathbf{p} = A\hat{\mathbf{x}} &= A(A^TA)^{-1}A^T\mathbf{b} \\ P &= A(A^TA)^{-1}A^T. \end{aligned}$$

Jeffrey Chasnov

If you understand Strang's proof,
Chasnov repeats it in the first 4 minutes.



Ref: https://www.youtube.com/watch?v=WABC6wmuLOk

Alex Townsend

Townsend's slide:
https://math.mit.edu/classes/18.085/summer2016/handouts/LeastSquares.pdf

Another way to show the same thing:

- https://theclevermachine.wordpress.com/2012/09/01/derivation-of-ols-normal-equations/
- https://sites.math.washington.edu/~burke/crs/308/LeastSquares.pdf

Online tool to perform differentiation of matrix:
http://www.matrixcalculus.org/

Matrix calculus reference (Advance):

- https://atmos.washington.edu/~dennis/MatrixCalculus.pdf
- https://www.comp.nus.edu.sg/~cs5240/lecture/matrix-differentiation.pdf
- https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf

NORMAL EQUATIONS:

$$A^T Ax = A^T b$$

Why the normal equations? To find out you will need to be slightly crazy and totally comfortable with calculus.

In general, we want to minimize[1]

$$f(x) = \|b - Ax\|_2^2 = (b - Ax)^T (b - Ax) = b^T b - x^T A^T b - b^T Ax + x^T A^T Ax.$$

If x is a global minimum of f, then its gradient $\nabla f(x)$ is the zero vector. Let's take the gradient of f remembering that

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}.$$

We have the following three gradients:

$$\nabla(x^T A^T b) = A^T b, \quad \nabla(b^T Ax) = A^T b, \quad \nabla(x^T A^T Ax) = 2A^T Ax.$$

To calculate these gradients, write out $x^T A^T b$, $b^T Ax$, and $x^T A^T Ax$, in terms of sums and differentiate with respect to $x_1, \ldots, x_n$ (this gets very messy).

Thus, we have

$$\nabla f(x) = 2A^T Ax - 2A^T b,$$

just like we saw in the example. We can solve $\nabla f(x) = 0$ or, equivalently $A^T Ax = A^T b$ to find the least squares solution. Magic.

Is this the global minimum? Could it be a maximum, a local minimum, or a saddle point? To find out we take the "second derivative" (known as the Hessian in this context):

$$Hf = 2A^T A.$$

Next week we will see that $A^T A$ is a positive semi-definite matrix and that this implies that the solution to $A^T Ax = A^T b$ is a global minimum of $f(x)$. Roughly speaking, $f(x)$ is a function that looks like a bowl.

Alexey Grigorev



Alexey Grigorev

A personal page

ML Wiki

Home Page Info

Normal Equation

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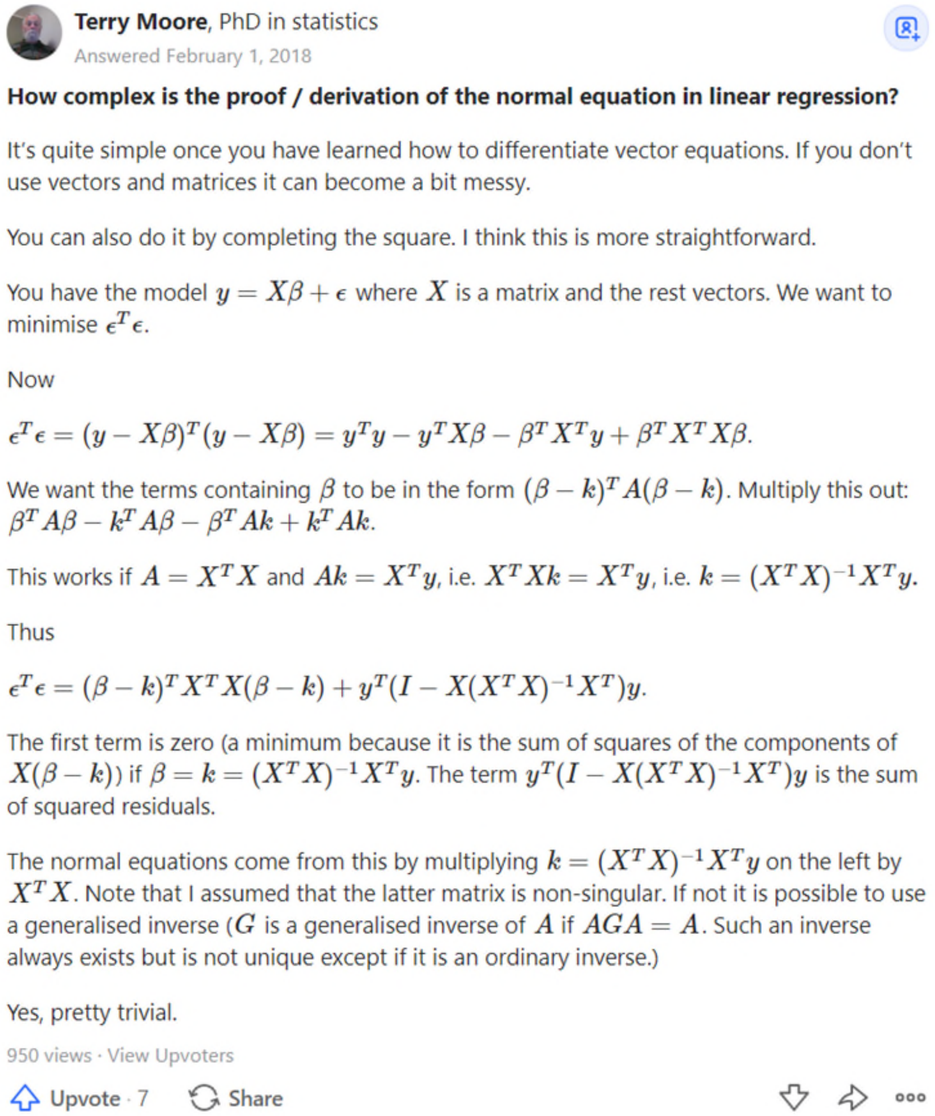
Projection onto Subspaces

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    4.2.1 Properties
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6 Projection onto Orthogonal Basis
7 Applications
8 Sources

Ref: http://mlwiki.org/index.php/Normal_Equation

Sometimes Quora has the Answer!



Terry Moore, PhD in statistics

Answered February 1, 2018

How complex is the proof / derivation of the normal equation in linear regression?

It's quite simple once you have learned how to differentiate vector equations. If you don't use vectors and matrices it can become a bit messy.

You can also do it by completing the square. I think this is more straightforward.

You have the model $y = X\beta + \epsilon$ where X is a matrix and the rest vectors. We want to minimise $\epsilon^T\epsilon$.

Now

$$\epsilon^T\epsilon = (y - X\beta)^T(y - X\beta) = y^Ty - y^TX\beta - \beta^TX^Ty + \beta^TX^TX\beta.$$

We want the terms containing β to be in the form $(\beta - k)^TA(\beta - k)$. Multiply this out: $\beta^TA\beta - k^TA\beta - \beta^TAk + k^TAk$.

This works if $A = X^TX$ and $Ak = X^Ty$, i.e. $X^TXk = X^Ty$, i.e. $k = (X^TX)^{-1}X^Ty$.

Thus

$$\epsilon^T\epsilon = (\beta - k)^TX^TX(\beta - k) + y^T(I - X(X^TX)^{-1}X^T)y.$$

The first term is zero (a minimum because it is the sum of squares of the components of $X(\beta - k)$) if $\beta = k = (X^TX)^{-1}X^Ty$. The term $y^T(I - X(X^TX)^{-1}X^T)y$ is the sum of squared residuals.

The normal equations come from this by multiplying $k = (X^TX)^{-1}X^Ty$ on the left by X^TX. Note that I assumed that the latter matrix is non-singular. If not it is possible to use a generalised inverse (G is a generalised inverse of A if $AGA = A$. Such an inverse always exists but is not unique except if it is an ordinary inverse.)

Yes, pretty trivial.

950 views · View Upvoters

Upvote · 7 Share

Ref: https://www.quora.com/How-complex-is-the-proof-derivation-of-the-normal-equation-in-linear-regression

Why the normal equation has a solution?

Does $A^T b$ lie in the column space of $A^T A$? If it does, why?

Asked 4 years, 11 months ago Modified 4 years, 11 months ago Viewed 729 times

1

This problem comes from the normal equations of linear least-squares:

$$(A^T A)x = A^T b$$

My question is whether or not $A^T b$ lies in the column space of $A^T A$ regardless of $A^T A$'s rank will be.

linear-algebra vector-spaces least-squares

Share Cite Edit Follow

asked Nov 8, 2017 at 3:45
Finley
993 ●1 ■5 ▲17

Add a comment

3 Answers

Sorted by: Highest score (default)

1

Yes, if A is a real $m \times n$ matrix and $b \in \mathbb{R}^m$, then $A^T b$ is in the column space of $A^T A$. Here is a proof:

Let $\hat{b}$ be the projection of b onto $R(A)$. Notice that $b - \hat{b}$ is orthogonal to every vector in $R(A)$, and in particular $b - \hat{b}$ is orthogonal to each column of A. This tells us that

$$A^T(b - \hat{b}) = 0. \qquad (\spadesuit)$$

Because $\hat{b} \in R(A)$, there exists x such that $Ax = \hat{b}$. Plugging in for $\hat{b}$ in equation $(\spadesuit)$, we see that $A^T Ax = A^T b$. This shows that $A^T b$ is in the range of A.

Share Cite Edit Follow

edited Nov 8, 2017 at 5:13

answered Nov 8, 2017 at 5:08
littleO
48.6k ●8 ■90 ▲157

Nice writeup by Micah Fogel – Illinois Maths and Science

Ref: http://staff.imsa.edu/~fogel/LinAlg/PDF/33%20Least%20Squares.pdf

Where he discussed 3 ways to find least squares: geometry, algebra and calculus

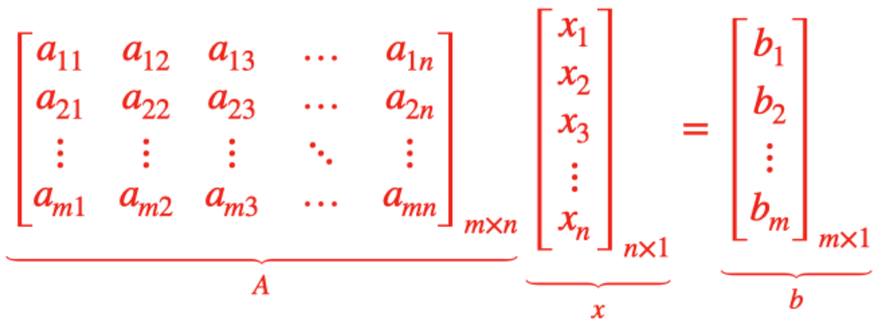
He discussed: Why is $A^TAx = A^Tb$ always solvable?

So why is $A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$ always solvable? Well, we use our Fundamental Theorem of Linear Algebra. The column space $\boldsymbol{C}(A^TA)$ is the orthogonal complement of the left nullspace of A^TA. Well, this is easier in symbols: $\boldsymbol{C}(A^TA) = \left(\boldsymbol{N}(A^TA)^T\right)^{\perp} = \left(\boldsymbol{N}(A^TA)\right)^{\perp} = \left(\boldsymbol{N}(A)\right)^{\perp} = \boldsymbol{C}(A^T)$ (we've seen that A and A^TA have the same nullspace because if $A\mathbf{x} = \mathbf{0}$ certainly $A^TA\mathbf{x} = \mathbf{0}$, but if $A^TA\mathbf{x} = \mathbf{0}$, we multiply on both sides by $\mathbf{x}^T$ and find the $\|A\mathbf{x}\| = 0$, so $A\mathbf{x} = \mathbf{0}$). But since the column spaces of A^TA and A^T are the same, and $A^T\mathbf{b}$ is in the column space of A^T we can certainly always solve $A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$.

He discussed: When A has dependent column , i.e (A^TA) $inverse$ does not exist

We know that if the columns of A are independent then A^TA is invertible, and there is a unique solution to this system. What if they aren't?

It turns out that we can still always solve the system, but now the solution won't be unique. We can add any null vector of A to a particular $\hat{\mathbf{x}}$ and obtain another solution. *This will be the point of the pseudoinverse*. We will pick out the particular solution that is in the row space of A. That way, any other solution will be this plus a null vector, which is orthogonal to the row space. So any other solution will be *longer* than the row space solution. Thus not only will the pseudoinverse make the error as small as possible, it will make the choice of $\hat{\mathbf{x}}$ as small as possible, too!



CX1104: **Linear Algebra for Computing**

Chap. No : **7.1.4**

Lecture : **Least Squares**

Topic : **Least Squares**

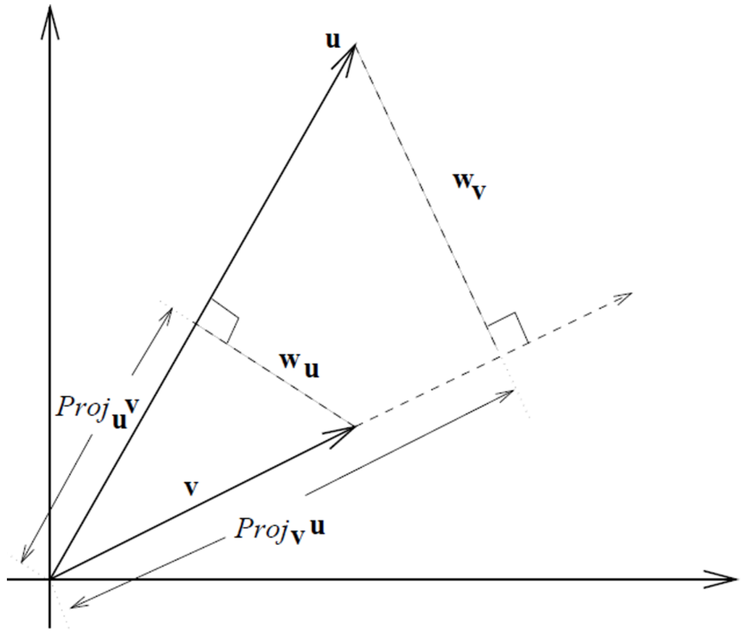
Concept : **Projection Matrix and its Properties**

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su**, **Vishal Choudhari**

Rev: 2nd July 2020

Projection matrix for a space spanned by a single vector v



$$Proj_{\mathbf{v}}\mathbf{u} = \mathbf{v}\left(\frac{\mathbf{u}^T\mathbf{v}}{||\mathbf{v}||^2}\right),$$

Ref: 6.2.2_Orthogonal_Projections

$$\begin{aligned} proj_{\mathbf{v}}\mathbf{u} &= \frac{\mathbf{v}\cdot\mathbf{u}}{\mathbf{v}\cdot\mathbf{v}}\,\mathbf{v} \\ &= \mathbf{v}\frac{\mathbf{v}\cdot\mathbf{u}}{\mathbf{v}\cdot\mathbf{v}} \end{aligned}$$

Next, use the transpose definition of the inner product followed by the associative property of multiplication. Remember, when performing the dot product, a scalar multiplier may be placed anywhere you wish.

$$\begin{aligned} proj_{\mathbf{v}}\mathbf{u} &= \frac{1}{\mathbf{v}^T\mathbf{v}}\,\mathbf{v}(\mathbf{v}^T\mathbf{u}) \\ &= \frac{1}{\mathbf{v}^T\mathbf{v}}\,(\mathbf{v}\mathbf{v}^T)\mathbf{u} \\ &= \frac{\mathbf{v}\mathbf{v}^T}{\mathbf{v}^T\mathbf{v}}\,\mathbf{u} \end{aligned}$$

The expression $\mathbf{v}\mathbf{v}^T$ is called an **outer product** (the transpose operator is outside the product versus its inside position in the inner product). If we define $P = \frac{\mathbf{v}\mathbf{v}^T}{\mathbf{v}^T\mathbf{v}}$, then the projection formula becomes

$$proj_{\mathbf{v}}\mathbf{u} = P\mathbf{u}, \text{ where } P = \frac{\mathbf{v}\mathbf{v}^T}{\mathbf{v}^T\mathbf{v}}.$$

The matrix P is called the projection matrix. You can project any vector onto the vector $\mathbf{v}$ by multiplying by the matrix P.

Ref: https://mse.redwoods.edu/darnold/math45/Activities/ProjectionMatrix/proj.pdf

Projection Matrix for col(A) and the Least Squares Solution

Consider solving the system of equations: $Ax = b$

- If b is not in the col(A), we find the least squares solution $\hat{x}$.

  $\hat{x}$ satisfies: $A^T A\hat{x} = A^T b$

  Thus, $\hat{x} = (A^T A)^{-1} A^T b$

Qn: **What is projection of b in the column space of A?**

Ans: $\hat{b} = proj_{Col(A)} b = A\hat{x}$

NOTE:

$$\hat{b} = proj_{Col(A)} b = A\hat{x}$$

$$\Longrightarrow \hat{b} = proj_{Col(A)} b = A(A^T A)^{-1} A^T b$$

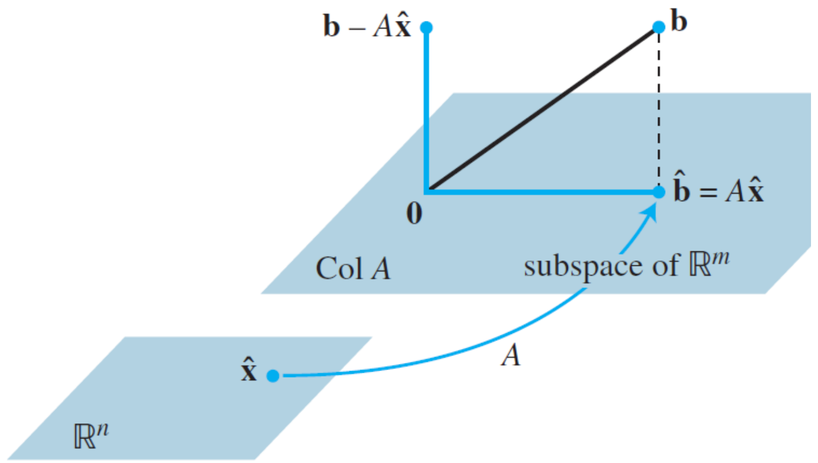


FIGURE 2 The least-squares solution $\hat{\mathbf{x}}$ is in $\mathbb{R}^n$.

$P = A(A^T A)^{-1} A^T$ is the **projection** matrix

The **vector b** can be **projected** into **column space of A** by **multiplying** it with **projection** matrix P.

Projection matrix P maps the actual response values b with predicted values $\hat{b}$.

Ref: https://en.wikipedia.org/wiki/Projection_matrix

Lay, Linear Algebra and its Applications (4th Edition)

360 CHAPTER 6 Orthogonality and Least Squares

Properties of Projection Matrix

When P is multiplied by vector b, the resulting vector $\hat{b} = Pb = A\hat{x}$ is the least squares (nearest) solution in the column space of A.

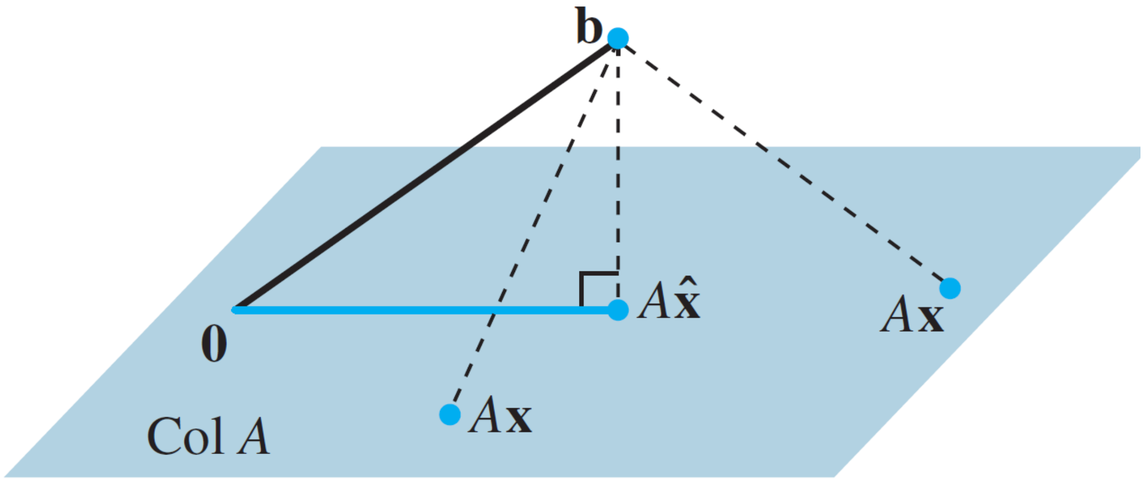


FIGURE 1 The vector **b** is closer to $A\hat{\mathbf{x}}$ than to $A\mathbf{x}$ for other $\mathbf{x}$.

Properties of Projection Matrix:

1. $P^T = P$
2. $P^N = P$ [Idempotent Property]

Lay, Linear Algebra and its Applications (4th Edition)

Properties of Projection Matrix

Lecture 16: Projection matrices and least squares

COURSE HOME

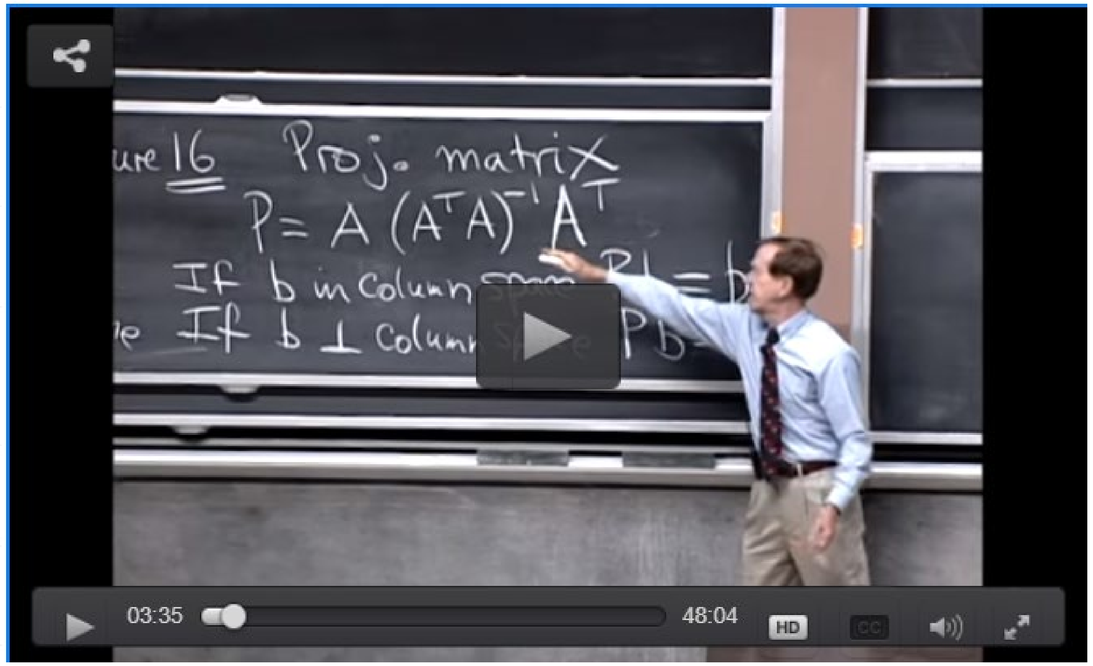
SYLLABUS

CALENDAR

INSTRUCTOR INSIGHTS

VIDEO LECTURES

READINGS



Ref: https://ocw.mit.edu/courses/mathematics/18-06-linear-algebra-spring-2010/video-lectures/lecture-16-projection-matrices-and-least-squares/

Properties of Projection Matrix:

1. $P^T = P$
2. $P^N = P$ [Idempotent Property]

Example 2: Projection Matrix Properties

4.3 *Projection matrices.* A matrix $P \in \mathbf{R}^{n\times n}$ is called a *projection matrix* if $P = P^T$ and $P^2 = P$.

(a) Show that if P is a projection matrix then so is $I - P$.

(b) Suppose that the columns of $U \in \mathbf{R}^{n\times k}$ are orthonormal. Show that UU^T is a projection matrix. (Later we will show that the converse is true: every projection matrix can be expressed as UU^T for some U with orthonormal columns.)

(c) Suppose $A \in \mathbf{R}^{n\times k}$ is full rank, with $k \leq n$. Show that $A(A^TA)^{-1}A^T$ is a projection matrix.

(d) If $S \subseteq \mathbf{R}^n$ and $x \in \mathbf{R}^n$, the point y in S closest to x is called the *projection of* x *on* S. Show that if P is a projection matrix, then $y = Px$ is the projection of x on $\mathcal{R}(P)$. (Which is why such matrices are called projection matrices . . .)

Ref: Boyd EE263 Homework 3, Q4.3:
https://web.stanford.edu/class/archive/ee/ee263/ee263.1082/hw/hw3sol.pdf

Example 1

Solution:

(a) To show that $I - P$ is a projection matrix we need to check two properties:

i. $I - P = (I - P)^T$

ii. $(I - P)^2 = I - P$.

The first one is easy: $(I - P)^T = I - P^T = I - P$ because $P = P^T$ (P is a projection matrix.) The show the second property we have

$$\begin{aligned}(I-P)^2 &= I - 2P + P^2 \\ &= I - 2P + P \qquad (\text{since } P = P^2) \\ &= I - P\end{aligned}$$

and we are done.

(b) Since the columns of U are orthonormal we have $U^TU = I$. Using this fact it is easy to prove that UU^T is a projection matrix, *i.e.*, $(UU^T)^T = UU^T$ and $(UU^T)^2 = UU^T$. Clearly, $(UU^T)^T = (U^T)^TU^T = UU^T$ and

$$\begin{aligned}(UU^T)^2 &= (UU^T)(UU^T) \\ &= U(U^TU)U^T \\ &= UU^T \qquad (\text{since } U^TU = I).\end{aligned}$$

(c) First note that $\left(A(A^TA)^{-1}A^T\right)^T = A(A^TA)^{-1}A^T$ because

$$\begin{aligned}\left(A(A^TA)^{-1}A^T\right)^T &= (A^T)^T\left((A^TA)^{-1}\right)^T A^T \\ &= A\left((A^TA)^T\right)^{-1}A^T \\ &= A(A^TA)^{-1}A^T.\end{aligned}$$

Also $\left(A(A^TA)^{-1}A^T\right)^2 = A(A^TA)^{-1}A^T$ because

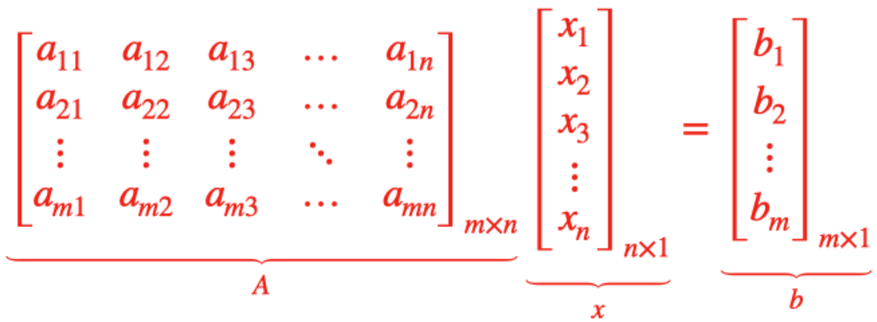
$$\begin{aligned}\left(A(A^TA)^{-1}A^T\right)^2 &= \left(A(A^TA)^{-1}A^T\right)\left(A(A^TA)^{-1}A^T\right) \\ &= A\left((A^TA)^{-1}A^TA\right)(A^TA)^{-1}A^T \\ &= A(A^TA)^{-1}A^T \qquad (\text{since } (A^TA)^{-1}A^TA = I).\end{aligned}$$

(d) To show that Px is the projection of x on $\mathcal{R}(P)$ we verify that the "error" $x - Px$ is orthogonal to *any* vector in $\mathcal{R}(P)$. Since $\mathcal{R}(P)$ is nothing but the span of the columns of P we only need to show that $x - Px$ is orthogonal to the columns of P, or in other words, $P^T(x - Px) = 0$. But

$$\begin{aligned}P^T(x-Px) &= P(x-Px) \qquad (\text{since } P = P^T) \\ &= Px - P^2x \\ &= 0 \qquad (\text{since } P^2 = P)\end{aligned}$$

and we are done.

Ref: Boyd EE263 Homework 3, Q4.3:
https://web.stanford.edu/class/archive/ee/ee263/ee263.1082/hw/hw3sol.pdf



CX1104: **Linear Algebra for Computing**

Chap. No : **7.1.5**

Lecture : **Least Squares**

Topic : **Least Squares**

Concept : **Summary Least Squares Solution**

Instructor: **A/P Chng Eng Siong**

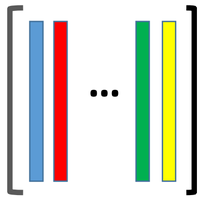
TAs: **Zhang Su**, **Vishal Choudhari**

Rev: 2nd July 2020

Solving Least Squares using QR Factorisation and MATLAB

Solving $Ax = b$:

Case 1



$M \approx N$

Say,

- A is square and invertible (full rank)
- then, b is in column space of A

Case 2

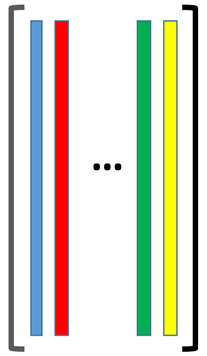


$M \ll N$

Under-determined

- As there are more unknowns than equations, infinitely many solutions exist.
- Hence, the goal then becomes to solve for x, such that, $||x||$ is minimised!

Case 3



$M \gg N$

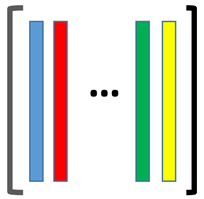
Over-determined

- b may not be in col space of A
- Hence, $b = Ax + \epsilon$
- ϵ models error/noise

Solving Least Squares using QR Factorisation and MATLAB

Solving $Ax = b$:

Case 1



$M \approx N$

Solution:

As b in column space of A, unique solution exists.

$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$, since A is square and has full rank.

Say,

- A is square and invertible (full rank)
- then, b is in column space of A

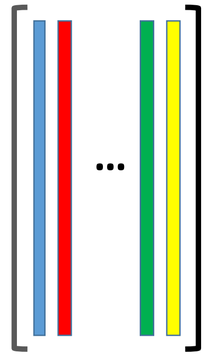
Solving Least Squares using QR Factorisation and MATLAB

For case 3, b is not in col. space of A.

Hence, an estimate of x, denoted by $\hat{x}$, such that least squares error ($||b - Ax||$) is minimised can be found.

Solving $Ax = b$:

Case 3



$M \gg N$

Over-determined

- b may not be in col space of A
- Hence, $b = Ax + \epsilon$
- ϵ models error/noise

Three ways to solve for case 3!

Way 1
Pseudoinverse

$\hat{x} = pinv(A) \times b$

Way 2
Inverting Normal Equation

$\hat{x} = (A^T A)^{-1} A^T b$

Way 3
QR

Let $A = QR$, where $Q^T Q = I$.

Hence, $Ax = b$ can be rewritten as:
$Q^T QRx = Q^T b$ (or)
$Rx = Q^T b$

Therefore, $\hat{x} = R^{-1} Q^T b$

Revisiting QR Factorisation and Solving $Ax = b$

Consider solving the system of equations: $Ax = b$

Through QR factorisation, A can be written as:

$$A = QR$$

where,

The Q Factor:

- Q is $m \times n$ with orthonormal columns $(Q^T Q = I)$
- If A is square $(m = n)$, then Q is orthogonal, i.e, $Q^T Q = QQ^T = I$

The R Factor:

- R is $n \times n$ upper triangular, with nonzero diagonal elements
- R is nonsingular (diagonal elements are nonzero)

So, $Ax = b$ can be rewritten as: $QRx = b$.

Multiplying both sides by Q^T yields:

$$Q^T QRx = Q^T b \text{ (or)}$$
$$Rx = Q^T b$$

Since R is an upper triangular matrix, x can be solved by:

1. Back-substitution
2. On MATLAB: $x = R^{-1} Q^T b$

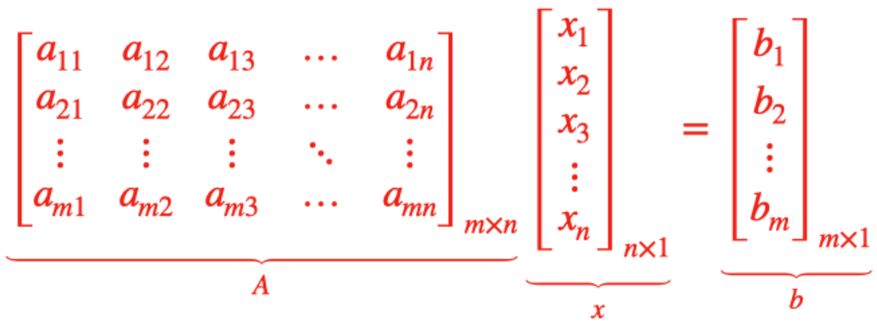
Algorithm Complexity

1. compute QR factorization $A = QR$ ($2mn^2$ flops if A is $m \times n$)
2. matrix-vector product $d = Q^T b$ ($2mn$ flops)
3. solve $Rx = d$ by back substitution (n^2 flops)

complexity: $2mn^2$ flops

Ref: “Why use QR to solve Ax=b?” by Dr. Peyam
https://www.youtube.com/watch?v=J41Ypt6Mftc





CX1104: **Linear Algebra for Computing**

Chap. No : **7.2.1**

Lecture : **Least Squares**

Topic : **Applications to Linear Models**

Concept : **Linear in the parameter Models**

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su**, **Vishal Choudhari**

Rev: 3 Oct 2021

Rev: 29th June 2020

Fitting a line to data

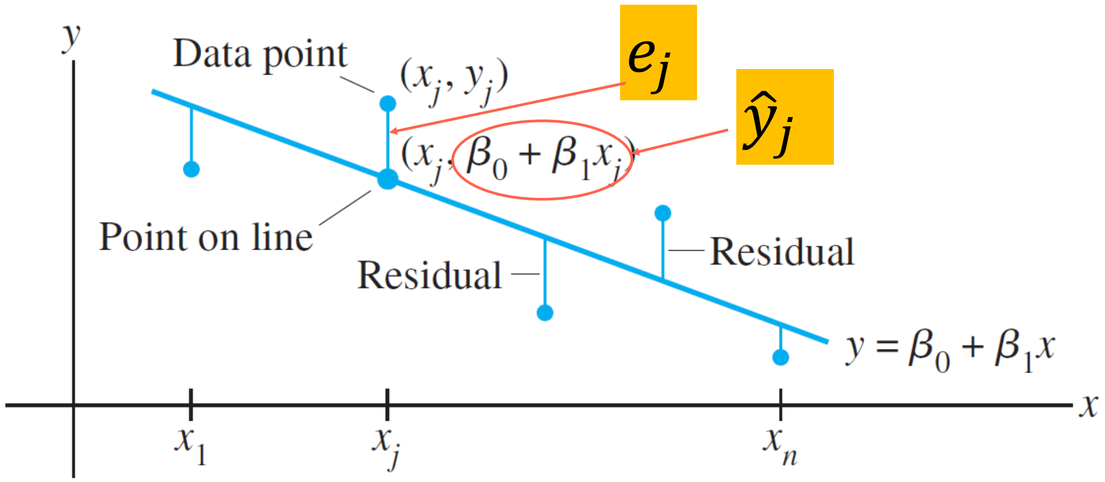


FIGURE 1 Fitting a line to experimental data.

We are given: $\{x_j, y_j\}$ where $j = 1..n$

Example

j	x_i	y_i
1	0.5	9.2
2	1.0	10.4
3	1.5	8.1

Find an equation $y = \beta_0 + \beta_1 x$ that describes the table.

Lay 5e, pg 371

Ref: Chasnov: https://www.youtube.com/watch?v=RlQBEhLhM8Y

Least-Squares Lines

The simplest relation between two variables x and y is the linear equation $y = \beta_0 + \beta_1 x$.[1] Experimental data often produce points $(x_1, y_1), \ldots, (x_n, y_n)$ that, when graphed, seem to lie close to a line. We want to determine the parameters β_0 and β_1 that make the line as "close" to the points as possible.

Suppose β_0 and β_1 are fixed, and consider the line $y = \beta_0 + \beta_1 x$ in Figure 1. Corresponding to each data point (x_j, y_j) there is a point $(x_j, \beta_0 + \beta_1 x_j)$ on the line with the same x-coordinate. We call y_j the *observed* value of y and $\beta_0 + \beta_1 x_j$ the *predicted* y-value (determined by the line). The difference between an observed y-value and a predicted y-value is called a *residual*.

[1] This notation is commonly used for least-squares lines instead of $y = mx + b$.

There are several ways to measure how "close" the line is to the data. The usual choice (primarily because the mathematical calculations are simple) is to add the squares of the residuals. The **least-squares line** is the line $y = \beta_0 + \beta_1 x$ that minimizes the sum of the squares of the residuals. This line is also called a **line of regression of y on x**, because any errors in the data are assumed to be only in the y-coordinates. The coefficients β_0, β_1 of the line are called (linear) **regression coefficients**.[2]

Actual $\boldsymbol{y}$ vector

Estimated $\boldsymbol{y}$ vector

residual vector $\boldsymbol{\epsilon}$, defined by $\boldsymbol{\epsilon} = \mathbf{y} - X\boldsymbol{\beta}$

$e_j = y_j - \hat{y}_j$, where $\hat{y}_j = estimated\ y_j$

Least Squares Solution to linear regression

If the data points were on the line, the parameters β_0 and β_1 would satisfy the equations

Predicted y-value		Observed y-value
$\beta_0 + \beta_1 x_1$	=	y_1
$\beta_0 + \beta_1 x_2$	=	y_2
$\vdots$		$\vdots$
$\beta_0 + \beta_1 x_n$	=	y_n

We can write this system as

$$X\boldsymbol{\beta} = \mathbf{y}, \quad \text{where } X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \tag{1}$$

Of course, if the data points don't lie on a line, then there are no parameters β_0, β_1 for which the predicted y-values in $X\boldsymbol{\beta}$ equal the observed y-values in $\mathbf{y}$, and $X\boldsymbol{\beta} = \mathbf{y}$ has no solution. This is a least-squares problem, $A\mathbf{x} = \mathbf{b}$, with different notation!

The square of the distance between the vectors $X\boldsymbol{\beta}$ and $\mathbf{y}$ is precisely the sum of the squares of the residuals. The $\boldsymbol{\beta}$ that minimizes this sum also minimizes the distance between $X\boldsymbol{\beta}$ and $\mathbf{y}$. *Computing the least-squares solution of* $X\boldsymbol{\beta} = \mathbf{y}$ *is equivalent to finding the* $\boldsymbol{\beta}$ *that determines the least-squares line in Figure 1.*

Least Squares Solution:

From

$$\boldsymbol{X\beta = y}$$

Then pre-multiply by $\boldsymbol{X^T}$ to get the normal equation,

$$\boldsymbol{X^T X\beta = X^T y}$$

Then premultiply by $(\boldsymbol{X^T X})^{-1}$,

$$(\boldsymbol{X^T X})^{-1}(\boldsymbol{X^T X})\boldsymbol{\beta} = \left(\boldsymbol{X^T X}\right)^{-1}\boldsymbol{X^T y}$$
$$\boldsymbol{\beta} = (\boldsymbol{X^T X})^{-1}\boldsymbol{X^T y}$$

Example 1

EXAMPLE 1 Find the equation $y = \beta_0 + \beta_1 x$ of the least-squares line that best fits the data points $(2, 1)$, $(5, 2)$, $(7, 3)$, and $(8, 3)$.

SOLUTION Use the x-coordinates of the data to build the design matrix X in (1) and the y-coordinates to build the observation vector $\mathbf{y}$:

$$X = \begin{bmatrix} 1 & 2 \\ 1 & 5 \\ 1 & 7 \\ 1 & 8 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 3 \end{bmatrix}$$

For the least-squares solution of $X\boldsymbol{\beta} = \mathbf{y}$, obtain the normal equations (with the new notation):

$$X^T X\boldsymbol{\beta} = X^T \mathbf{y}$$

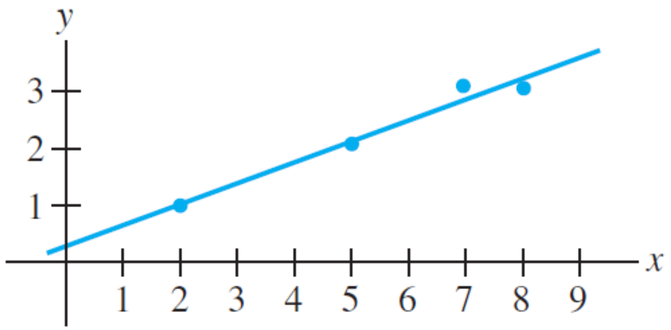


FIGURE 2 The least-squares line $y = \frac{2}{7} + \frac{5}{14}x$.

That is, compute

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 5 & 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 5 \\ 1 & 7 \\ 1 & 8 \end{bmatrix} = \begin{bmatrix} 4 & 22 \\ 22 & 142 \end{bmatrix}$$

$$X^T \mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 5 & 7 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 9 \\ 57 \end{bmatrix}$$

The normal equations are

$$\begin{bmatrix} 4 & 22 \\ 22 & 142 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 9 \\ 57 \end{bmatrix}$$

Hence

$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 4 & 22 \\ 22 & 142 \end{bmatrix}^{-1} \begin{bmatrix} 9 \\ 57 \end{bmatrix} = \frac{1}{84} \begin{bmatrix} 142 & -22 \\ -22 & 4 \end{bmatrix} \begin{bmatrix} 9 \\ 57 \end{bmatrix} = \frac{1}{84} \begin{bmatrix} 24 \\ 30 \end{bmatrix} = \begin{bmatrix} 2/7 \\ 5/14 \end{bmatrix}$$

Thus the least-squares line has the equation

$$y = \frac{2}{7} + \frac{5}{14}x$$

Lay 5e, pg 372

Least Squares Fit to Other Curves

Least-Squares Fitting of Other Curves

When data points $(x_1, y_1), \ldots, (x_n, y_n)$ on a scatter plot do not lie close to any line, it may be appropriate to postulate some other functional relationship between x and y.

The next two examples show how to fit data by curves that have the general form

$$y = \beta_0 f_0(x) + \beta_1 f_1(x) + \cdots + \beta_k f_k(x) \qquad (2)$$

where $f_0, \ldots, f_k$ are known functions and $\beta_0, \ldots, \beta_k$ are parameters that must be determined. As we will see, equation (2) describes a linear model because it is linear in the unknown parameters.

Linear in the parameter model:

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where

residual vector $\boldsymbol{\epsilon}$, defined by $\boldsymbol{\epsilon} = \mathbf{y} - X\boldsymbol{\beta}$

Any equation of this form is referred to as a **linear model**. Once X and $\mathbf{y}$ are determined, the goal is to minimize the length of $\boldsymbol{\epsilon}$, which amounts to finding a least-squares solution of $X\boldsymbol{\beta} = \mathbf{y}$. In each case, the least-squares solution $\hat{\boldsymbol{\beta}}$ is a solution of the normal equations

$$X^T X\boldsymbol{\beta} = X^T\mathbf{y}$$

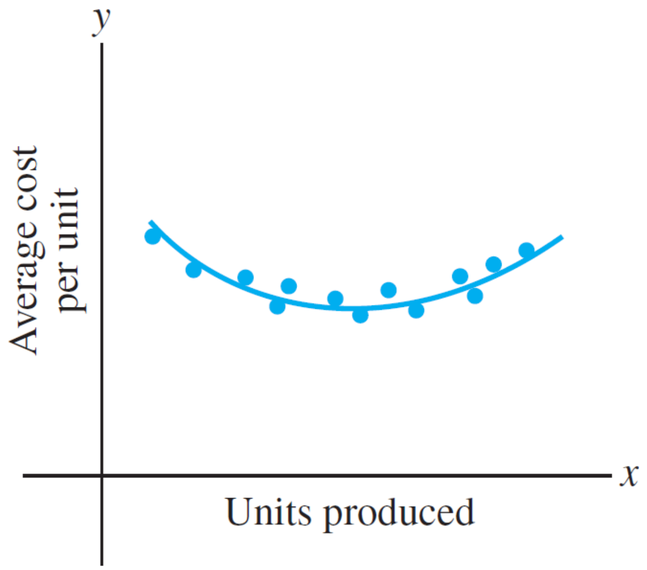


FIGURE 3

Average cost curve.

Ref1: https://datascience.stackexchange.com/questions/12274/what-does-linear-in-parameters-mean

Example 2: fitting to curves

EXAMPLE 2 Suppose data points $(x_1, y_1), \ldots, (x_n, y_n)$ appear to lie along some sort of parabola instead of a straight line. For instance, if the x-coordinate denotes the production level for a company, and y denotes the average cost per unit of operating at a level of x units per day, then a typical average cost curve looks like a parabola that opens upward (Figure 3). In ecology, a parabolic curve that opens downward is used to model the net primary production of nutrients in a plant, as a function of the surface area of the foliage (Figure 4). Suppose we wish to approximate the data by an equation of the form

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 \tag{3}$$

Describe the linear model that produces a "least-squares fit" of the data by equation (3).

SOLUTION Equation (3) describes the ideal relationship. Suppose the actual values of the parameters are $\beta_0, \beta_1, \beta_2$. Then the coordinates of the first data point (x_1, y_1) satisfy an equation of the form

$$y_1 = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \epsilon_1$$

where ϵ_1 is the residual error between the observed value y_1 and the predicted y-value $\beta_0 + \beta_1 x_1 + \beta_2 x_1^2$. Each data point determines a similar equation:

$$\begin{aligned} y_1 &= \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \epsilon_1 \\ y_2 &= \beta_0 + \beta_1 x_2 + \beta_2 x_2^2 + \epsilon_2 \\ &\vdots \qquad\qquad\quad \vdots \\ y_n &= \beta_0 + \beta_1 x_n + \beta_2 x_n^2 + \epsilon_n \end{aligned}$$

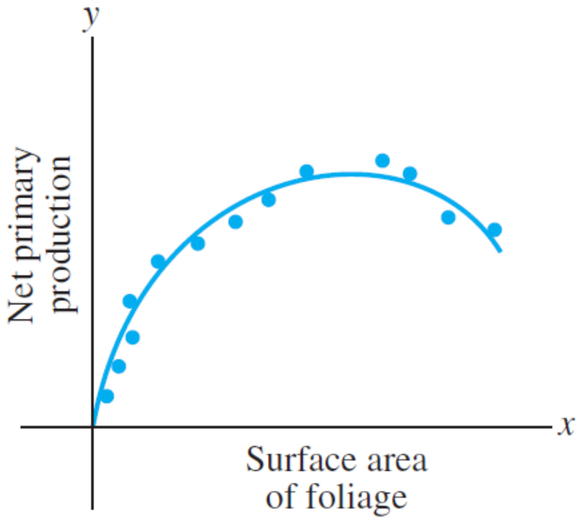


FIGURE 4
Production of nutrients.

It is a simple matter to write this system of equations in the form $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$. To find X, inspect the first few rows of the system and look for the pattern.

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$$\mathbf{y} \quad = \quad X \quad\quad \boldsymbol{\beta} \quad + \quad \boldsymbol{\epsilon}$$

Observation vector — Design matrix — Parameter vector — Residual vector

■

Lay 5e, pg 373

Maths/LA/Tut7
Least Squares

16 Nov 2020

CES

Last updated: 19 Oct 2021

Tutorial 7 Help links

Youtube link: playlist

https://www.youtube.com/playlist?list=PLki3aFwg-9exa_oECiSJtTtaei7zTKwbI

PDF

Q1-6: https://www.dropbox.com/s/pc33morjzp3fmxm/Tut7_Q1_6_ces.pdf?dl=0

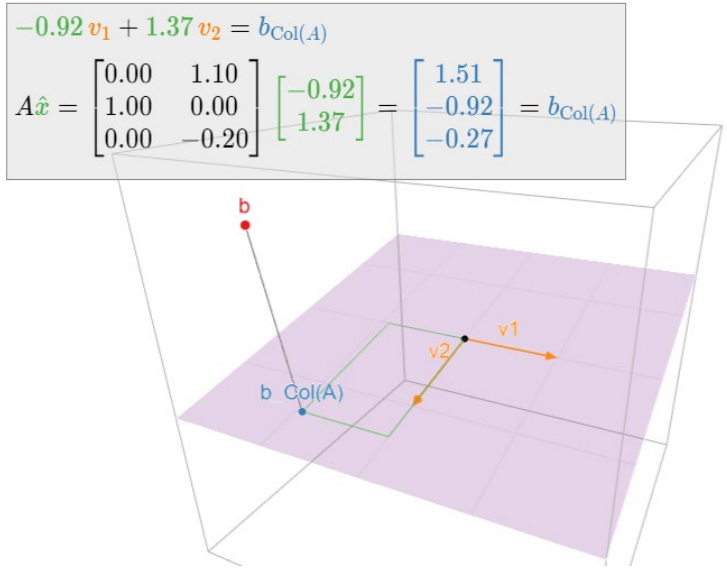
Overview of Least Squares

1) Cornell's CS3220 class:

https://www.cs.cornell.edu/~bindel/class/cs3220-s12/notes/lec10.pdf

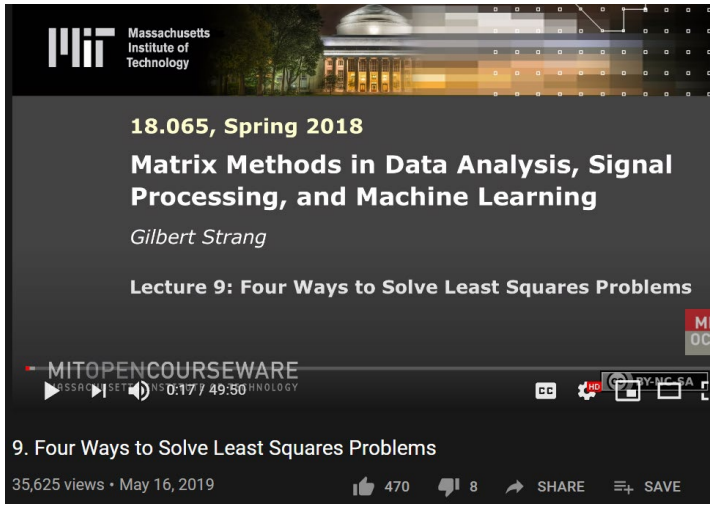
2) GaTech's online book with nice visualization applet

https://textbooks.math.gatech.edu/ila/least-squares.html



How many ways to solve the least squares

- There are several ways to solve the least squares solution.
- See:
  - https://stats.stackexchange.com/questions/160179/do-we-need-gradient-descent-to-find-the-coefficients-of-a-linear-regression-mode/164164#164164
  - Strang's 4 ways to solve the least squares (Advance):
    - https://www.youtube.com/watch?v=ZUU57Q3CFOU



Why is $A^T A$ invertible when A has full col rank (related to Q5-17e NS 5-18d)

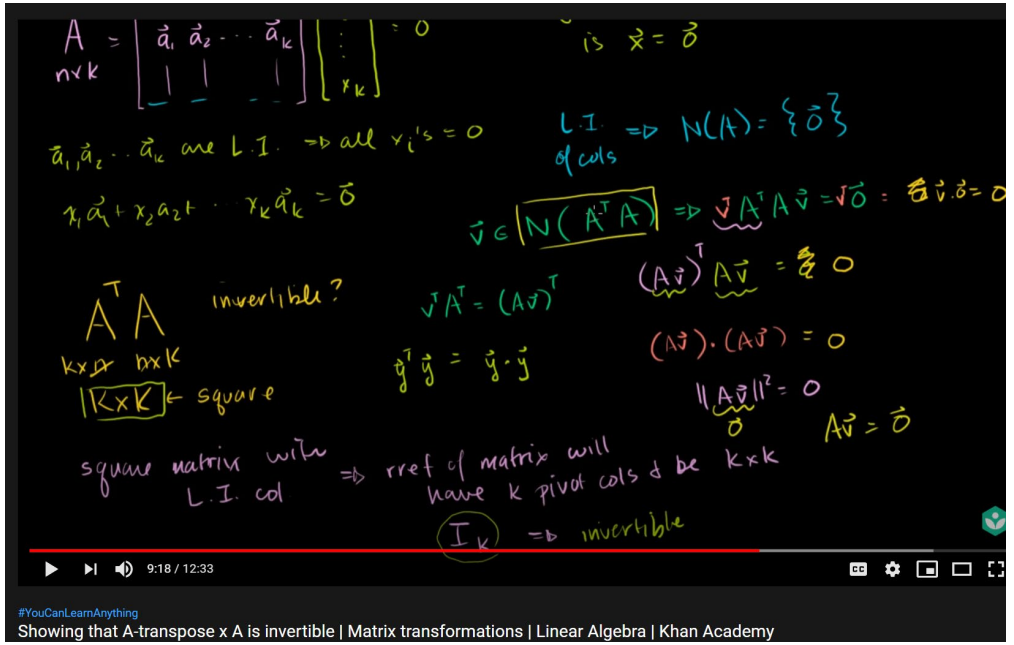
1) Khan Academy:

https://www.youtube.com/watch?v=ESSMQH6Y5OA

2) See rank of $A^T A$ vs A

https://math.stackexchange.com/questions/349738/prove-operatornamerankata-operatornameranka-for-any-a-in-m-m-times-n

3) To proof that rank(A) == rank(A^TA)
So that inverse (A^TA) exist, so that least squares is unique when the columns of A Is independent
https://yutsumura.com/rank-and-nullity-of-a-matrix-nullity-of-transpose/



Proof of Tut 7/Q6

The reader may have noticed that we have been careful to say "the least-squares solutions" in the plural, and "a least-squares solution" using the indefinite article. This is because a least-squares solution need not be unique: indeed, if the columns of A are linearly dependent, then $Ax = b_{\mathrm{Col}(A)}$ has infinitely many solutions. The following theorem, which gives equivalent criteria for uniqueness, is an analogue of this corollary in Section 6.3.

Theorem. *Let A be an $m \times n$ matrix and let b be a vector in $\mathbf{R}^m$. The following are equivalent:*

1. *$Ax = b$ has a unique least-squares solution.*
2. *The columns of A are linearly independent.*
3. *A^TA is invertible.*

In this case, the least-squares solution is

$$\widehat{x} = (A^TA)^{-1}A^Tb.$$

Proof. ^

The set of least-squares solutions of $Ax = b$ is the solution set of the consistent equation $A^TAx = A^Tb$, which is a translate of the solution set of the homogeneous equation $A^TAx = 0$. Since A^TA is a square matrix, the equivalence of 1 and 3 follows from the invertible matrix theorem in Section 5.1. The set of least squares-solutions is also the solution set of the consistent equation $Ax = b_{\mathrm{Col}(A)}$, which has a unique solution if and only if the columns of A are linearly independent by this important note in Section 2.5.

Important Note 2.5.9 (Recipe: Checking linear independence).

A set of vectors $\{v_1, v_2, \ldots, v_k\}$ is linearly independent if and only if the vector equation

$$x_1v_1 + x_2v_2 + \cdots + x_kv_k = 0$$

has only the trivial solution, if and only if the matrix equation $Ax = 0$ has only the trivial solution, where A is the matrix with columns $v_1, v_2, \ldots, v_k$:

$$A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_k \\ | & | & & | \end{pmatrix}.$$

This is true if and only if A has a pivot position in every column. Solving the matrix equatiion $Ax = 0$ will either verify that the columns $v_1, v_2, \ldots, v_k$ are linearly independent, or will produce a linear dependence relation by substituting any nonzero values for the free variables.

Ref:
https://textbooks.math.gatech.edu/ila/1553/least-squares.html

Uniqueness of least squares solution ONLY when the columns of A are independent

- https://courses.math.tufts.edu/math70/Section%20Summaries/Chapter6/sect%206.5.pdf

Theorem 14 Let A be an $m \times n$ matrix. The following statements are logically equivalent.
(a) The equation $A\mathbf{x} = \mathbf{b}$ has a unique least squares solution for each $\mathbf{b}$ in $\mathbb{R}^n$.
(b) The columns of A are linearly independent.
(c) The matrix A^TA is invertible.
When these statements are true, the least squares solution $\hat{\mathbf{x}}$ is given by:

$$\hat{\mathbf{x}} = (A^TA)^{-1}A^T\mathbf{b}.$$

(This provides a fast solution method when $(A^TA)^{-1}$ is easy to find.)

Proof of Theorem 14:

$a \to b$ Recall that we proved (or will prove) that $\text{Nul}(A) = \text{Nul}(A^TA)$. Suppose the columns of A are linearly independent. Then $\text{Nul}(A) = \{\mathbf{0}\}$. Since $\text{Nul}(A) = \text{Nul}(A^TA)$, $\text{Nul}(A^TA) = \{\mathbf{0}\}$ also. This means that $(A^TA)\mathbf{x} = \mathbf{0} \to \mathbf{x} = \mathbf{0}$. Thus the columns of A^TA are linearly independent, and since A^TA is a square matrix, $A^TA \sim I$ so A^TA is invertible.

$b \to c$ Suppose the $n \times n$ matrix A^TA is invertible. We can always find least squares solutions using the normal equations:

$$A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$$

But since A^TA is invertible we can apply the inverse to both sides of the previous equation to get:

$$(A^TA)^{-1}(A^TA)\hat{\mathbf{x}} = (A^TA)^{-1}A^T\mathbf{b}$$

$$I\hat{\mathbf{x}} = (A^TA)^{-1}A^T\mathbf{b}$$

$$\hat{\mathbf{x}} = (A^TA)^{-1}A^T\mathbf{b}$$

This is a unique solution.

$c \to a$ Suppose the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for all $\mathbf{b} \in \mathbb{R}^m$. Then since $\mathbf{0}_{\mathbb{R}^m} \in \mathbb{R}^m$, $A\mathbf{x} = \mathbf{0}_{\mathbb{R}^m}$ has a unique solution. Since $\mathbf{x} = \mathbf{0}_{\mathbb{R}^n}$ *is* a solution, it must be the only one. Thus $A\mathbf{x} = \mathbf{0} \to \mathbf{x} = \mathbf{0}$ and so the columns of A are linearly independent.

Infinitely many solutions for least squares (When col of A are dependent)

Example (Infinitely many least-squares solutions).

Find the least-squares solutions of $Ax = b$ where:

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & 2 & -3 \end{pmatrix} \qquad b = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}.$$

Solution

We have

$$A^T A = \begin{pmatrix} 3 & 3 & -3 \\ 3 & 5 & -7 \\ -3 & -7 & 11 \end{pmatrix} \qquad A^T b = \begin{pmatrix} 6 \\ 0 \\ 6 \end{pmatrix}.$$

We form an augmented matrix and row reduce:

$$\left(\begin{array}{ccc|c} 3 & 3 & -3 & 6 \\ 3 & 5 & -7 & 0 \\ -3 & -7 & 11 & 6 \end{array}\right) \xrightarrow{\text{RREF}} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 5 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

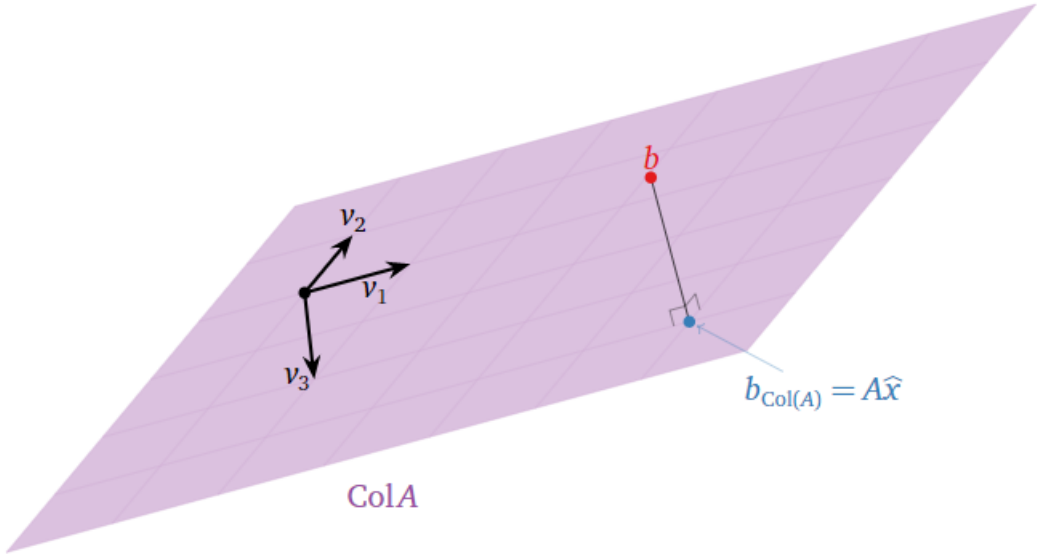
The free variable is x_3, so the solution set is

$$\begin{cases} x_1 = -x_3 + 5 \\ x_2 = 2x_3 - 3 \\ x_3 = x_3 \end{cases} \xrightarrow[\text{vector form}]{\text{parametric}} \widehat{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 5 \\ -3 \\ 0 \end{pmatrix}.$$

For example, taking $x_3 = 0$ and $x_3 = 1$ gives the least-squares solutions

$$\widehat{x} = \begin{pmatrix} 5 \\ -3 \\ 0 \end{pmatrix} \quad \text{and} \quad \widehat{x} = \begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}.$$

Geometrically, we see that the columns v_1, v_2, v_3 of A are coplanar:



Therefore, there are many ways of writing $b_{\text{Col}(A)}$ as a linear combination of v_1, v_2, v_3.

https://textbooks.math.gatech.edu/ila/1553/least-squares.html

Is $A^T b$ in the column space of A?

Corollary:

1) Why does $A^T Ax = A^T b$
(the normal equation)

always have a solution?

Ref:

http://staff.imsa.edu/~fogel/LinAlg/PDF/33%20Least%20Squares.pdf

Let's ask how close we can come to solving the equation. $\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} r \\ h \end{bmatrix}$ is guaranteed to be in the column space of the matrix. So instead of using the real **b**, let's find the thing in the column space that is as close to **b** as possible, and solve for that instead! Let **p** be the projection of **b** into the column space. Then the error vector $\mathbf{e} = \mathbf{b} - \mathbf{p}$ is as small as possible. Let's call the solution to this new problem $\hat{\mathbf{x}}$ so we are solving $A\hat{\mathbf{x}} = \mathbf{p}$. The one thing we know about **e** is that it is orthogonal to the column space, so it is in the left nullspace. That is, $A^T\mathbf{e} = \mathbf{0}$. This means that $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$, or $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$.

So instead of $A\mathbf{x} = \mathbf{b}$, we solve the *normal equations* $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$. (We will show later that this *always* has a solution). In this case, we multiply both sides by $A^T = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}$ to obtain the system $\begin{bmatrix} 14 & 6 \\ 6 & 3 \end{bmatrix}\begin{bmatrix} \hat{r} \\ \hat{h} \end{bmatrix} = \begin{bmatrix} 31 \\ 14 \end{bmatrix}$. A little elimination shows that $\hat{r} = 3/2$ and $\hat{h} = 5/3$.

So we guess our little plant started our 5/3 cm tall and grew at a rate of 3/2 cm/day. This is clearly wrong, since it would predict heights of 19/6, 14/3, and 37/6 instead of 3, 5,and 6, so we're off by –1/6, 1/3, and –1/6 respectively. The is, $\mathbf{e} = (-1/6, 1/3, -1/6)$.

So why is $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$ always solvable? Well, we use our Fundamental Theorem of Linear Algebra. The column space $\boldsymbol{C}(A^T A)$ is the orthogonal complement of the left nullspace of $A^T A$. Well, this is easier in symbols: $\boldsymbol{C}(A^T A) = \left(\boldsymbol{N}(A^T A)^T\right)^{\perp} = \left(\boldsymbol{N}(A^T A)\right)^{\perp} = \left(\boldsymbol{N}(A)\right)^{\perp} = \boldsymbol{C}(A^T)$ (we've seen that A and $A^T A$ have the same nullspace because if $A\mathbf{x} = \mathbf{0}$ certainly $A^T A\mathbf{x} = \mathbf{0}$, but if $A^T A\mathbf{x} = \mathbf{0}$, we multiply on both sides by $\mathbf{x}^T$ and find the $||A\mathbf{x}|| = 0$, so $A\mathbf{x} = \mathbf{0}$). But since the column spaces of $A^T A$ and A^T are the same, and $A^T\mathbf{b}$ is in the column space of A^T we can certainly always solve $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$.

Another explanation: why $A^TAc = A^Tx$ is consistent

Ref:
https://textbooks.math.gatech.edu/ila/projections.html#projections-ATA-formula

Theorem. *Let A be an $m \times n$ matrix, let $W = \mathrm{Col}(A)$, and let x be a vector in $\mathbf{R}^m$. Then the matrix equation*

$$A^TAc = A^Tx$$

in the unknown vector c is consistent, and x_W is equal to Ac for any solution c.

Proof. Let $x = x_W + x_{W^\perp}$ be the orthogonal decomposition with respect to W. By definition x_W lies in $W = \mathrm{Col}(A)$ and so there is a vector c in $\mathbf{R}^n$ with $Ac = x_W$. Choose any such vector c. We know that $x - x_W = x - Ac$ lies in $W^\perp$, which is equal to $\mathrm{Nul}(A^T)$ by this important note in Section 6.2. We thus have

$$0 = A^T(x - Ac) = A^Tx - A^TAc$$

and so

$$A^TAc = A^Tx.$$

This exactly means that $A^TAc = A^Tx$ is consistent. If c is any solution to $A^TAc = A^Tx$ then by reversing the above logic, we conclude that $x_W = Ac$.

Related: space of A^TA vs A

- https://math.stackexchange.com/questions/1272572/row-space-and-column-space-of-at-a-and-a-at

2

If $Ax = 0$, then $A^TAx = 0$, which means $N(A) \subset N(A^TA)$, $N(A)$ is the null space of A.

On the other hand, if $A^TAx = 0$, then

$$x^TA^TAx = 0, \quad \text{or} \quad \|Ax\| = 0$$

which means $Ax = 0$, and thus

$$N(A^TA) \subset N(A) \quad \text{and} \quad N(A^TA) = N(A)$$

Since $\text{rank}(A) = n - N(A)$, there is

$$\text{rank}(A) = \text{rank}(A^TA)$$

Suppose $A = [\alpha_1, \cdots, \alpha_n]$ (α_i is the column vector of A), then

$$A^TA = A^T[\alpha_1, \cdots, \alpha_n] = [A^T\alpha_1, \cdots, A^T\alpha_n]$$

For each column of A^TA

$$\begin{aligned} A^T\alpha_i &= [\beta_1 \cdots \beta_n]\alpha_i \\ &\quad (\beta_i \text{ is the column of } A^T \text{ and row of } A) \\ &= [\beta_1 \cdots \beta_n] \begin{bmatrix} a_{i1} \\ \vdots \\ a_{in} \end{bmatrix} \\ &= \sum_{j=1}^{n} a_{ij}\beta_j \end{aligned}$$

So column of A^TA is the linear combination of rows of A, or

$$\text{col}(A^TA) = \text{row}(A)$$

Obviously $\text{rank}(A^T) = \text{rank}(A)$, so

$$\text{row}(A^TA) = \text{col}(A^TA) = \text{row}(A)$$

Similarly we have

$$\text{row}(AA^T) = \text{col}(AA^T) = \text{row}(A^T) = \text{col}(A)$$

Nullity of A and A^TA

- https://yutsumura.com/rank-and-nullity-of-a-matrix-nullity-of-transpose/

Problem 140

Let A be an $m \times n$ matrix. The nullspace of A is denoted by $\mathcal{N}(A)$.
The dimension of the nullspace of A is called the nullity of A.
Prove the followings.

(a) $\mathcal{N}(A) = \mathcal{N}(A^{\mathrm{T}}A)$.

(b) $\mathrm{rank}(A) = \mathrm{rank}(A^{\mathrm{T}}A)$.

Proof.

(a) $\mathcal{N}(A) = \mathcal{N}(A^{\mathrm{T}}A)$.

Show $\mathcal{N}(A) \subset \mathcal{N}(A^{\mathrm{T}}A)$

Consider any $\mathbf{x} \in \mathcal{N}(A)$. Then we have $A\mathbf{x} = \mathbf{0}$. Multiplying it by A^{T} from the left, we obtain

$$A^{\mathrm{T}}A\mathbf{x} = A^{\mathrm{T}}\mathbf{0} = \mathbf{0}.$$

Thus $\mathbf{x} \in \mathcal{N}(A^{\mathrm{T}}A)$, and hence $\mathcal{N}(A) \subset \mathcal{N}(A^{\mathrm{T}}A)$.

Show $\mathcal{N}(A) \supset \mathcal{N}(A^{\mathrm{T}}A)$

On the other hand, let $\mathbf{x} \in \mathcal{N}(A^{\mathrm{T}}A)$. Thus we have

$$A^{\mathrm{T}}A\mathbf{x} = \mathbf{0}.$$

Multiplying it by $\mathbf{x}^{\mathrm{T}}$ from the left, we obtain

$$\mathbf{x}^{\mathrm{T}}A^{\mathrm{T}}A\mathbf{x} = \mathbf{x}^{\mathrm{T}}\mathbf{0} = \mathbf{0}.$$

This implies that we have

$$\mathbf{0} = (A\mathbf{x})^{\mathrm{T}}(A\mathbf{x}) = ||A\mathbf{x}||^2$$

and the length of the vector $A\mathbf{x}$ is zero, thus the vector $A\mathbf{x} = \mathbf{0}$. Hence $\mathbf{x} \in \mathcal{N}(A)$, and we obtain $\mathcal{N}(A) \supset \mathcal{N}(A^{\mathrm{T}}A)$.

(b) $\mathrm{rank}(A) = \mathrm{rank}(A^{\mathrm{T}}A)$

We use the rank-nullity theorem and obtain

$$\mathrm{rank}(A) = n - \dim(\mathcal{N}(A)) = n - \dim(\mathcal{N}(A^{\mathrm{T}}A)) = \mathrm{rank}(A^{\mathrm{T}}A).$$

(Note that the size of the matrix $A^{\mathrm{T}}A$ is $n \times n$.)

Rank of A and $A^T A$ are same

- https://math.stackexchange.com/questions/349738/prove-operatornamerankata-operatornameranka-for-any-a-in-m-m-times-n

- And therefore if A is tall and full rank, then $A^T A$ is invertible

Method1: Using dimension and rank

127

Let $\mathbf{x} \in N(A)$ where $N(A)$ is the null space of A.

So,

$$\begin{aligned} & A\mathbf{x} = \mathbf{0} \\ \implies & A^T A\mathbf{x} = \mathbf{0} \\ \implies & \mathbf{x} \in N(A^T A) \end{aligned}$$

Hence $N(A) \subseteq N(A^T A)$.

Again let $\mathbf{x} \in N(A^T A)$

So,

$$\begin{aligned} & A^T A\mathbf{x} = \mathbf{0} \\ \implies & \mathbf{x}^T A^T A\mathbf{x} = \mathbf{0} \\ \implies & (A\mathbf{x})^T (A\mathbf{x}) = \mathbf{0} \\ \implies & A\mathbf{x} = \mathbf{0} \\ \implies & \mathbf{x} \in N(A) \end{aligned}$$

Hence $N(A^T A) \subseteq N(A)$.

Therefore

$$\begin{aligned} & N(A^T A) = N(A) \\ \implies & \dim(N(A^T A)) = \dim(N(A)) \\ \implies & \operatorname{rank}(A^T A) = \operatorname{rank}(A) \end{aligned}$$

Method2: Using SVD

Let r be the rank of $A \in \mathbb{R}^{m \times n}$. We then have the SVD of A as

$$A_{m \times n} = U_{m \times r} \Sigma_{r \times r} V^T_{r \times n}$$

This gives $A^T A$ as

$$A^T A = V_{n \times r} \Sigma^2_{r \times r} V^T_{r \times n}$$

which is nothing but the SVD of $A^T A$. From this it is clear that $A^T A$ also has rank r. In fact the singular values of $A^T A$ are nothing but the square of the singular values of A.

share cite improve this answer follow

answered Apr 3 '13 at 3:25

user17762

Why QR is more stable? (related to Q5-18f)

See explanation in slide 8.16- 8.17 in

- http://www.seas.ucla.edu/~vandenbe/133A/lectures/ls.pdf

Projection Matrix Proof

$$P = A(A^T A)^{-1} A^T$$

Assume that A has full column rank, (all columns independent)
Show that above P has two properties

a) $P = P^T$
b) $PP = P$

Pre-amples:

1) We need to proof $(A^{-1})^T = (A^T)^{-1}$

- see: https://math.stackexchange.com/questions/340233/transpose-of-inverse-vs-inverse-of-transpose

6

I would derive the formula step by step this way.

Lets have invertible matrix A, so you can write following equation (definition of inverse matrix):

$$AA^{-1} = I$$

Lets transpose both sides of equation. (using $I^T = I$, $(XY)^T = Y^T X^T$)

$$(AA^{-1})^T = I^T$$

$$(A^{-1})^T A^T = I$$

From the last equation we can say (based on the definition of inverse matrix) that A^T is inverse of $(A^{-1})^T$. So we can write following.

$$(A^{-1})^T)^{-1} = A^T$$

By inverting both sides of equation we obtain the desired formula.

$$(A^{-1})^T = (A^T)^{-1}$$